The Value of AI-Generated Metadata for UGC Platforms: Evidence from a Large-Scale Field Experiment

Xinyi Zhang[1], Chenshuo Sun[2], Renyu Zhang[3], Khim-Yong Goh[4]
[1] National University of Singapore, Singapore, xinyizhang@u.nus.edu
[2] Zhongguancun Academy, Beijing, China, csun@stern.nyu.edu
[3] The Chinese University of Hong Kong, Hong Kong, China, philipzhang@cuhk.edu.hk
[4] National University of Singapore, Singapore, gohky@comp.nus.edu.sg

Generative AI (GAI) is increasingly used to produce user-facing content, yet its value for content *metadata*, the less prominently displayed content on user-generated (UGC) platforms, remains unclear. We address this gap through a large-scale field experiment on a leading short-video platform in Asia, where about one million producers were given suggestions of AI-generated video titles which are often not populated by producers themselves. Producers exhibited three responses: rejection (3.2%), direct adoption (59.5%), and human-AI co-creation (37.3%; in terms of context revision and emotion revision). The provision of AI-generated titles significantly boosted content consumption, increasing valid watches by 1.6% and watch duration by 0.9%. Mechanism analysis showed that AI-generated metadata improved user-video matching in the platform's recommender system. Decomposing these effects reveals a *coverage effect* through which AI alleviated metadata sparsity, and a *quality effect* whereby human-AI co-created titles outperformed both AI-generated and human-written titles. Further analysis shows that both context revision and emotion revision improved content consumption: they raised valid watches by 22.4% and 21.3%, and watch duration by 20.4% and 21.8%, respectively, and were associated with richer linguistic attributes. In sum, our results highlight the value of AI-generated metadata and human-AI co-creation in enhancing user-content matching and content consumption on UGC platforms. This study contributes theoretically to research on GAI, recommender systems, and platform operations, and offers practical insights for platform managers to design more engaging UGC platforms.

Key words: Generative AI, Video metadata, User-video matching, Short-video platforms, Human-AI Co-creation

1. Introduction

Generative AI (GAI) and AI-generated content (AIGC) have demonstrated significant value and potential across industries by efficiently producing high-quality content, such as generating advertising copy, improving customer service, and enhancing media production.[1] These GAI

[1] https://www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/how-generative-ai-can-boost-consumer-marketing.

tools help streamline content creation, reduce manual effort, and improve output quality. Market research[2] shows that after adopting GAI, 75% of marketers reported higher content production, while 79% observed improvements in content quality. Such an efficiency boost has also fueled the growth of the GAI market, which was valued at USD 11.6 billion in 2023 and is projected to grow to USD 175.3 billion by 2033, with a compound annual growth rate of 31.2%.

On user-generated content (UGC) platforms, particularly short-video platforms like TikTok, content metadata such as titles and tags typically includes descriptive information about the content. This metadata provides structured details such as themes and subjects that may allow recommender systems to better categorize and understand content (Wei et al. 2024). Research on YouTube (Hoiles et al. 2017) has shown that improved video metadata was associated with a 25% increase in video search visibility. A Nielsen report indicates that books with complete metadata had 2.2 times higher sales than those with incomplete metadata.[3]

Despite its potential value, metadata is largely sparse on online platforms. Malik and Tian (2017) observe that only 42,000 (i.e., 0.26%) of 1.6 million YouTube videos had complete metadata. Similarly, a dataset of over 50,000 IMDb movies showed that less than 25% had complete metadata.[4] Other platforms, such as Spotify[5] and TikTok,[6] have reported similar challenges. Metadata sparsity is primarily driven by the time-consuming nature of metadata generation. Additionally, because metadata is less visible to users and, hence, does not directly engage them, producers often undervalue its importance (Peng et al. 2023). Platforms, however, may not want to mandate metadata generation, as doing so could discourage content uploads and ultimately harm the platform ecosystem.[7]

The recent emergence of GAI technologies has provided UGC platforms with new solutions to the metadata sparsity challenge. Video metadata generation is a typical video-to-text application within the AIGC ecosystem, where algorithms automatically convert video inputs into textual descriptors such as titles and hashtags, thereby reducing the burden on content creators (see Table A.1). In practice, AI-generated video metadata tools are widely available for online platforms (see Table A.2), raising several questions of both academic and practical interest.

AI-generated metadata can fundamentally change how metadata is created on UGC platforms. Instead of being purely human-generated, metadata may now emerge from a dynamic interplay

[2] https://market.us/report/generative-ai-in-content-creation-market/.

[3] https://pnmais.com/wp-content/uploads/2023/11/Report_NielsenBookData_MVB_Metadata_Frankfurt_2022.pdf.

[4] https://www.kaggle.com/datasets/rajugc/imdb-movies-dataset-based-on-genre?resource=download.

[5] https://soundcharts.com/blog/music-metadata.

[6] A dataset of 1,729 TikTok videos indicated that 45% of short videos have missing hashtags. See https://www.kaggle.com/datasets/vbradculbertson/tiktok-trending-metadata?select=sug_users_vids1.csv.

[7] https://www.multicollab.com/blog/user-generated-content/.

between human producers and AI suggestions. This shift raises our first research question (RQ1): *How do content producers interact with AI-generated metadata?* Existing GAI studies (Chen and Chan 2023, Su et al. 2024) largely examine AIGC (e.g., advertising creatives) that engage content viewers, where producers' revisions aim to enhance persuasiveness and user appeal. In contrast, metadata is not the primary user-facing content and is generated to support algorithmic interpretation and recommendation (Agrawal et al. 2023). This distinction implies that producers' incentives and behaviors when interacting with AI-generated metadata are likely to diverge from those observed in prior AIGC research, yet remain empirically underexplored.

Another important question faced by UGC platforms gives rise to our second research question (RQ2): *What value does AI-generated metadata bring to a UGC platform?* While AI holds promise in generating detailed and relevant metadata (Agrawal et al. 2023, Wei et al. 2024), the answer to this question is not yet conclusive. As discussed above, metadata is instrumental for user-content matching for UGC platforms, but no causal evidence has been reported in the literature to quantify its downstream effects on content consumption in real-world settings. Furthermore, AI-generated metadata could introduce new challenges, such as hallucination, misinformation, stereotypes, and biased responses (Liang et al. 2021, Susarla et al. 2023). Hence, AI may generate misleading metadata that misrepresents the content, resulting in poor user-content matching and disengaging the matched users. Therefore, rigorous empirical research is needed to quantify the actual value of AI-generated metadata for UGC platforms. Prior GAI studies (Chen and Chan 2023, Su et al. 2024) have focused on user-facing AIGC, with the influence of AI-generated metadata mostly overlooked. Similarly, data augmentation studies (Wei et al. 2024, Ellis et al. 2018) have largely overlooked the potential of GAI to generate video metadata.

With a good understanding of AI-generated metadata's value, our third research question (RQ3) asks: *How can UGC platforms best leverage AI-generated metadata to boost content consumption?* One important consideration is whether platforms should deploy AI-generated metadata as a full substitute for human-generated metadata or as a complementary tool that supports producers. Current GAI research (Chen and Chan 2023, Zhou and Lee 2024) has yet to provide conclusive evidence on which approach is more effective. In terms of quantity, AI-generated metadata may improve metadata fulfillment by lowering the effort required for metadata generation, but it may also hinder metadata fulfillment by imposing additional verification and evaluation costs on producers. In terms of quality, AI may outperform humans by learning from large datasets and identifying patterns overlooked by humans (Chen and Chan 2023), yet it may also struggle with context-specific cases (Longoni et al. 2019, Granulo et al. 2021), where human insights about content goals and intended audiences are important for creating context- and emotion-rich metadata (Sun et al. 2022). Hence, it remains unclear whether AI-generated metadata,

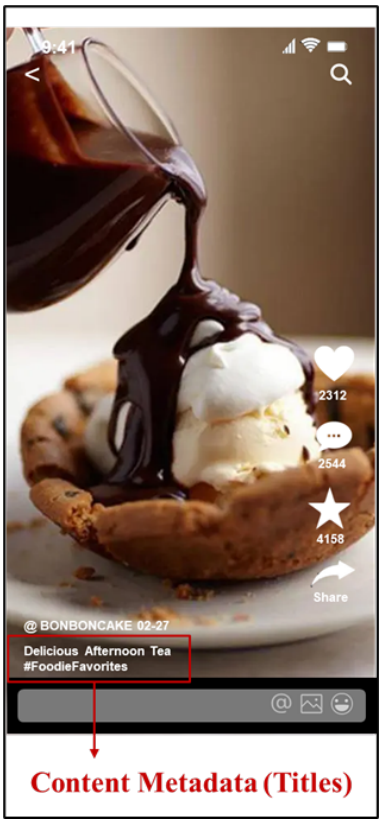


Figure 1 **Viewer Interface for Watching Videos on Platform A**

human-generated metadata, or a co-creation of both human and AI is most effective in increasing content consumption on UGC platforms. The answer is important for platforms' operational decisions on deploying AI-generated metadata for producers' UGC production process.

To answer these questions, we collaborated with a short-video platform in Asia (hereafter "Platform A") to conduct a large-scale randomized field experiment. Users on Platform A primarily consume videos recommended by the platform. Video titles are the metadata we focus on and could often be overlooked (see Figure 1 where video titles are small and positioned at the bottom left corner of the viewers' interface[8]). This metadata is a crucial part of the video profile data used in the recommender system that matches users with relevant videos on Platform A. However, like many UGC platforms, Platform A faces the challenge of metadata sparsity. In our dataset (see Section 3.3 for details), only 60.7% of videos had titles during the pre-treatment period. To address this challenge, Platform A developed a generative AI tool capable of producing video titles directly from video frame data.

In our experiment, users were randomly assigned to either the treatment or the control condition. Video producers in the treatment group received access to AI-generated titles on the video posting

[8] A similar example can be seen on social media platforms like Instagram, where titles and tags are often hidden and require users to click "expand" to view them. Similarly, on e-commerce platforms like Amazon or Taobao, product specifications are often tucked away in drop-down menus, making them less visible unless being actively searched for.

page, and had the flexibility to either delete, amend, or fully adopt the AI-generated title, whereas producers in the control group could only write titles themselves. Our study covered the treatment period from August 8th to 21st, 2023 and included 2,048,033 producers, each of whom posted at least one video. Therefore, our study provides unique large-scale experimental evidence to causally examine the impact of GAI tools on a real-world platform, whereas earlier works were mostly based on observational data or lab-based methods.

The analyses of our experimental data yield several important insights. **As our first contribution, we identify how producers interacted with AI-generated titles.** Producers exhibited three interaction patterns: rejection (3.2%), direct adoption (59.5%), and human-AI co-creation (37.3%), where co-creation involved either context revision (21.0%) or emotion revision (16.3%). Access to AI-generated titles affected producers' titling behaviors, but neither the quantity nor the efficiency of video production. Finally, video content and producer characteristics shaped which interaction pattern producers followed. For example, producers of utilitarian-content videos were more likely to directly adopt AI-generated titles, whereas those of hedonic-content videos were more likely to engage in human-AI co-creation through emotion revision.

Our second contribution is to quantify the value of AI-generated metadata to the platform. We find that the access to AI-generated video titles significantly boosted video consumption, increasing the number of valid watches[9] by 1.6%, and watch duration by 0.9%. As video titles on Platform A constitute an important data input of the recommender system, our analyses show that AI-generated titles enhanced consumption primarily by improving the user-video matching accuracy of the recommender system.

Our third contribution is to offer actionable insights on how platforms could best leverage AI-generated metadata to boost content consumption. We find that the impact of AI-generated titles can be decomposed into a *coverage effect* and a *quality effect*, corresponding to the quantity and quality dimensions emphasized in prior GAI research. The coverage effect captures whether AI-generated titles mitigate *metadata sparsity*. We find that access to AI-generated titles increased the likelihood of a video having a title by 41.4% and tags by 72.4%. The quality effect reflects the impact of AI-generated titles on viewership outcomes relative to human-generated titles, thereby isolating the effect of title quality rather than coverage. Our analyses find that among titled videos, access to AI-generated titles reduced viewership, with declines of 37.9% in valid watches and 32.6% in watch duration, suggesting that AI-generated titles were generally of lower quality than human-generated ones. However, when producers co-created with AI by substantially revising

[9] At the viewer-video level, a watch qualifies if the viewer watches the entire video for videos between 3 and less than 7 seconds, and at least 7 seconds for videos of 7 seconds or longer. In our paper, valid watch is a video-level measure defined as the number of such watches a video receives.

AI-generated titles (as indicated by lower textual similarity between the original AI-generated titles and the final producer-adopted ones), viewership improved significantly. Specifically, each 10% decrease in this similarity increased valid watches by 9.8% and watch duration by 9.2%. Lower similarity was associated with richer linguistic attributes. These findings underscore the value of human-AI co-creation in improving metadata quality.

To further understand how human-AI co-creation influences content consumption, we next distinguish between two specific co-creation modes, i.e., context revision and emotion revision. Results show that both modes significantly improved viewership outcomes. Context revision increased valid watches by 22.4% and watch duration by 20.4%, while emotion revision increased valid watches by 21.3% and watch duration by 21.8%. In contrast, directly adopting AI-generated titles without revision reduced valid watches by 49.4% and watch duration by 41.9%. Together, these results indicate that although AI-generated titles underperformed human-generated ones, human input in the form of context and emotion revision substantially enhanced title quality and boosted viewership. Moreover, both forms of human-AI co-creation were associated with richer linguistic attributes, suggesting that human revisions produced richer metadata. Complementary qualitative evidence supports this interpretation. Open-ended responses from 1,925 treatment producers consistently pointed to an "inspiration effect" where AI-generated titles served as creative prompts that motivated producers to craft richer (e.g., lexically richer) titles. Therefore, our analyses show that human-AI co-creation is not merely a byproduct of metadata provision but a central mechanism through which producers enhance the relevance of AI-generated titles.

The rest of the paper proceeds as follows. Section 2 reviews related literature. Section 3 details our field setting, experimental design, and data. Section 4 presents producer interactions. Section 5 examines the economic impact of AI-generated metadata on content consumption, and Section 6 analyzes underlying mechanisms. Section 7 provides additional analyses and robustness checks. Section 8 concludes the paper and discusses practical implications and future research directions.

2. Literature Review

Our paper speaks to four streams of literature: (1) GAI and its collaborations with humans, (2) data augmentation and recommender system, (3) platform operations, and (4) content attributes and user engagement.

2.1. GAI and its Collaborations with Humans

Emerging studies have examined its economic impact across various fields including labor market (Liu et al. 2024), firm innovation (Cheng et al. 2022), marketing (Reisenbichler et al. 2022), artwork (Zhou and Lee 2024), and knowledge sharing (Burtch et al. 2023). A central dimension of this stream is how GAI affects both the quantity (also called frequency or productivity) and quality of content

production. On the quantity side, GAI can boost content production by lowering production costs (Chen and Chan 2023, Reisenbichler et al. 2022). However, GAI may also raise verification and evaluation costs, which can ultimately reduce content production (Boyacı et al. 2024, Becker et al. 2025). On the quality side, some studies report quality improvements (Reisenbichler et al. 2022), whereas others observe quality declines due to reduced novelty or the exit of expert producers (Burtch et al. 2023, Chen and Chan 2023, Zhou and Lee 2024).

Our contribution to this literature is fourfold. First, we contribute to the growing literature on GAI and content production by examining how producers interact with AI-generated metadata, a form of AIGC that is fundamentally distinct from the user-facing content studied in prior work. Existing studies largely focus on AIGC such as advertising creatives or AI-generated summaries that directly engage viewers, where producer revisions primarily aim to enhance persuasiveness, readability, and user appeal (Chen and Chan 2023, Su et al. 2024, Reisenbichler et al. 2022). In contrast, metadata (e.g., video titles) is generated primarily to support algorithmic interpretation and recommendation rather than direct user consumption (Agrawal et al. 2023). This distinction implies different producer incentives when interacting with AI-generated metadata, leading to revision behaviors that may diverge systematically from those documented in prior AIGC research. We contribute by uncovering how producers interact with and revise AI-generated metadata.

Second, the nascent literature on GAI and user engagement has largely focused on settings where AIGC directly interacts with consumers. For example, AI-generated summaries are displayed on product pages, reducing search costs and directly shaping consumer behavior (Su et al. 2024), while AI-generated advertising copy directly influences purchasing decisions (Chen and Chan 2023, Reisenbichler et al. 2022). These studies primarily attribute improved user responses to enhancements in content quality. In contrast, we examine a distinct mechanism through which GAI can improve user responses by enhancing user-content matching via augmented metadata within recommender systems, a channel that has received little empirical attention. Using a unique experimental design, we study AI-generated metadata that is rarely visible to users and does not directly engage them, allowing us to attribute changes in content consumption primarily to improved matching rather than user engagement. We thus extend the mechanism from enhanced content quality to improved matching accuracy through reduced metadata sparsity. Moreover, existing studies offer ambiguous predictions regarding metadata generation: while AI may lower production costs and increase metadata coverage (Reisenbichler et al. 2022), it may also raise confirmation and evaluation costs that constrain metadata fulfillment (Becker et al. 2025), and the quality of AI-generated metadata may either exceed or fall short of human-generated alternatives (Chen and Chan 2023, Reisenbichler et al. 2022). Our study disentangles these competing forces by

separating the coverage effect (increased metadata fulfillment) from the quality effect (improved contextual, emotional, and lexical richness of metadata).

Third and method-wise, most prior GAI studies conduct laboratory experiments (Chen and Chan 2023) or treat the launch of ChatGPT or other large language model tools as an event and leverage its timing for econometric identification (usually with Difference-in-Differences method) (e.g., Zhou and Lee 2024, Burtch et al. 2023). Our study enhances these studies by using a randomized field experiment to causally identify the impact of GAI tool availability. This experiment provides the exogenous variation in the input of AI-generated metadata in the platform's recommender system and thus allows us to identify its causal effect on consumption outcomes in a real-world setting.

Fourth, our research adds to the growing literature on human-AI collaboration, which has primarily shown that combining human input with AI tools outperforms both full automation and human-only approaches. This has been explored in studies on non-GAI tools (Anthony et al. 2023, Boyacı et al. 2024) and more recent work on GAI tools (Chen and Chan 2023, Zhou and Lee 2024, Wang et al. 2025). Research on GAI tools has identified two primary modes of content co-creation: human-revised AI-generated content and AI-revised human-generated content (Chen and Chan 2023). These studies often compare the linguistic or visual attributes of the content generated by AI, humans, and their co-creation, to understand how these different approaches impact content consumption (Chen and Chan 2023, Zhou and Lee 2024, Reisenbichler et al. 2022). However, because the AIGC examined in these studies is prominently shown to users, their findings mainly identify which linguistic or visual attributes make content more appealing and engaging to users. These studies are less informative about which attributes of human-AI co-created metadata improve user-content matching in recommender systems. For example, attributes such as emojis may increase engagement by making content more appealing or arousing to users, but their complex symbolic nature may make them less useful for recommender systems in inferring content meaning and matching content with relevant users. We speak to this open question by studying AI-generated metadata, a form of AIGC whose value lies less in directly engaging users than in helping recommender systems better match content with users. This setting allows us to provide evidence on how producers revise AI-generated titles and how such human modifications improve user-content matching and, in turn, video viewership.

2.2. Data Augmentation and Recommender System

Recommender system studies have documented that challenges in user-generated video metadata, such as noise, sparsity, and incompleteness, inhibit accurate user-content matching (Wei et al. 2024). To address these issues, data augmentation techniques are developed to enhance data quality. Solutions for handling data sparsity or missing data include, e.g., fuzziness methods (Choi

et al. 2018), imputation via supervised learning (Ellis et al. 2018), active feature-value acquisition (Saar-Tsechansky et al. 2009), and Monte Carlo likelihood estimation (Peng et al. 2023).

We extend this body of literature in three ways. First, prior research has largely overlooked the potential of GAI to enhance metadata in recommender systems, and even when such attempts exist, they primarily focus on algorithmic aspects without validation in real-world settings (e.g., Agrawal et al. 2023, Wei et al. 2024). Our study contributes by providing large-scale experimental evidence to quantify the economic value of AI-generated metadata in improving content consumption. Second, existing studies have not distinguished between two mechanisms through which GAI improves recommender system performance: coverage effect (increased metadata fulfillment) and the quality effect (improved contextual, emotional, and lexical richness of metadata). Our study contributes by disentangling these two effects in the context of AI-generated metadata. These findings provide new insights into recommender system design. Third, unlike past studies that retroactively impute missing data through algorithmic estimation, we propose a proactive approach that enables producers to access AI-generated metadata during the metadata generation process.

2.3. Platform Operations

Our research extends the growing literature that addresses operational problems on online platforms. This literature has examined how to build effective systems for pricing (Cui et al. 2022, Zhang et al. 2022), review systems (Cui et al. 2020a), logistic systems (Bai et al. 2022a), social fairness (Wang et al. 2023, Clyde et al. 2024), content production (Zeng et al. 2023), advertisement delivery (Ye et al. 2023), and content consumption (Fang et al. 2023). It has also studied how to ensure service quality (Cui et al. 2020b) and participants' responses to platform interventions (Lysyakov and Viswanathan 2023, Bai et al. 2022b).

We contribute to this stream of research in two ways. First, our work enriches the research that seeks to improve consumers' content consumption on UGC platforms. Prior studies have explored consumer-side interventions such as content prioritization (Dukes and Liu 2024) and personalized content distribution (Wei et al. 2024), as well as producer-side interventions such as financial incentives (Kuang et al. 2019), performance feedback (Huang et al. 2019), social norms (Burtch et al. 2018), and more recently, AI or GAI tools (He et al. 2021). The producer-side interventions largely focus on improving content quality to boost content consumption. Our research takes this literature a step further by examining the impact of a new operational intervention (i.e., AI-generated metadata) on content consumption outcomes.

Second, our work speaks to the emerging literature that empirically tests information-based interventions for solving operational problems for online platforms. Examples of interventions include providing producers with more information about customers (Buell et al. 2017), services or

products (Kesavan and Kushwaha 2020, Sun et al. 2022), and competitors (Cui et al. 2020a, Zeng et al. 2023). These interventions primarily enhance consumer engagement by improving content production in terms of speed, capacity, and quality. Our study extends this literature by introducing a new type of information-based intervention, i.e., AI-generated metadata that leverages metadata generation to drive content consumption by enhancing user-content matching.

2.4. Content Attributes and User Engagement

Our study is also closely related to the literature on how content attributes shape user engagement. Prior studies have systematically examined how a variety of content attributes affect engagement, including emotional valence and arousal (Berger and Milkman 2012), linguistic style and narrativity (Berger et al. 2023), content length and partitioning (Lee et al. 2025), and headline framing features such as concreteness and curiosity (Banerjee and Urminsky 2025). Beyond textual features, prior research has also investigated image- (Li and Xie 2020) and video-level attributes (Guo et al. 2024).

We contribute by examining a different type of textual content, i.e., video metadata (titles) on short-video platforms. Prior research has largely focused on prominently displayed textual attributes that directly capture users' attention, such as news articles (Berger and Milkman 2012, Berger et al. 2023), e-series books (Lee et al. 2025), and article headlines (Banerjee and Urminsky 2025). In contrast, metadata on short-video platforms serves a different role. Rather than functioning primarily as a direct cue for viewers, it helps producers describe content and helps recommender systems interpret and match videos with viewers. Our study advances the understanding of how metadata attributes enhance user-content matching.

3. Field Setting, Experimental Design, and Data

3.1. Research Context

We collaborated with one of the largest short-video platforms in Asia (Platform A), which has over 300 million daily active users. Like TikTok, users on the platform can be either content producers or viewers. Producers post short videos on Platform A to enhance viewership and/or engagement and attract new followers, aiming to increase advertising opportunities and revenue. Users visit Platform A either to be entertained by videos that catch their interest (organic browse) or to search for specific videos related to a topic (search-oriented browse).

Viewers can consume videos and engage with others for free on Platform A. Beyond viewing the videos uploaded by the producers, viewers can also like videos, leave comments, forward content both on and beyond Platform A, and follow producers for long-term video consumption and engagement. The platform generates revenue primarily through online advertising, i.e., disseminating advertising videos to viewers. Therefore, accurately matching the content with viewers to improve video consumption and engagement is crucial to Platform A's business model.

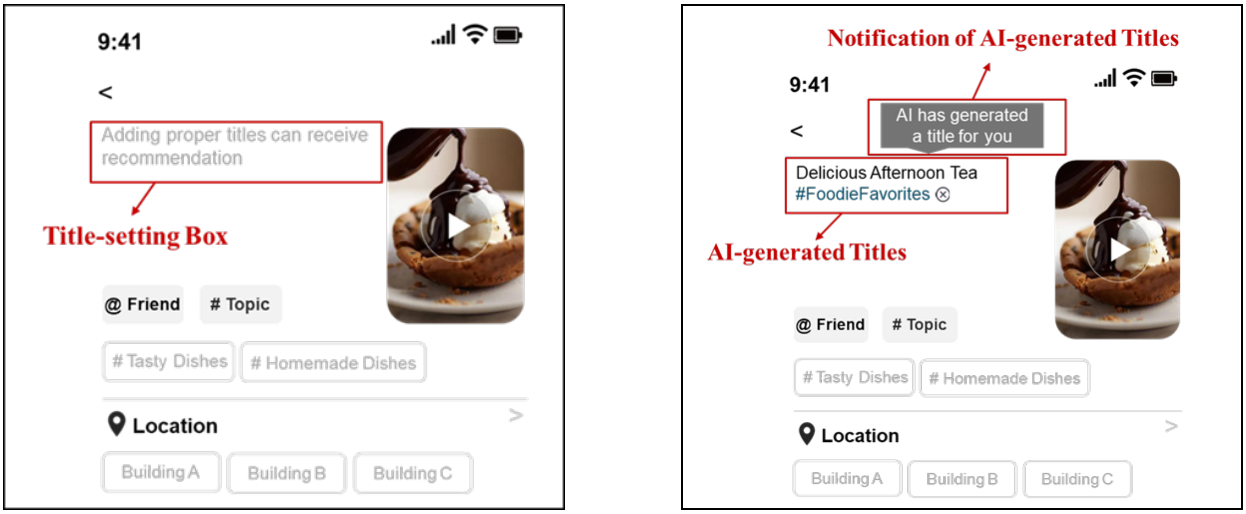
Content on Platform A is distributed through two primary channels: organic recommendations and search-query-oriented recommendations. Organic recommendations generate a personalized video feed based on a user's viewing history and preferences, catering to those browsing without active searches. Search-query-based recommendations, on the other hand, respond directly to users' search queries. The video title, including hashtags and descriptions, is the only content metadata used to improve the relevance of content recommendations.

Platform A's recommender system functions in two stages (see Figure B.1): candidate generation and ranking (Davidson et al. 2010). Both stages rely on four types of data inputs: (1) video profile, (2) user profile, (3) user engagement data (e.g., watch count, duration, likes), and (4) search queries (if any). Video profile data include both technical metadata (e.g., file format and upload date) and content metadata (i.e., titles), along with the producer profile and extracted features from processing raw video streams (e.g., video category). User profile data include user features (e.g., age and gender) and device features (e.g., device model). In the candidate generation stage, algorithms such as content-based collaborative filtering and context-aware methods are used to select relevant video candidates based on user input. A common approach is to pick videos that are closely related to the ones previously watched by the user, utilizing techniques like co-visitation counts and matrix factorization. The ranking stage then takes these candidates and prioritizes them based on the likelihood of user engagement. When posting videos, producers are encouraged to submit titles with hashtags or descriptions (see Figure 2(a)). There is no word limit for titles, and a producer can also post a video without a title. Allowing this flexibility helps maintain high upload rates, as mandatory title requirements could complicate the posting process and discourage content production.[10]

Video content metadata (e.g., title) is a crucial part of video profile information in Platform A's recommender system. Unlike technical metadata (e.g., upload timestamp), which is automatically logged by Platform A, a title requires user input. Although the platform can still recommend videos using other inputs, titles provide more specific context than the broad insights generated by extracted features, such as visual patterns, audio cues, and category classifications.[11] These features, while useful, often lack the precise contextual information required for accurate recommendations. In contrast, video titles offer structured and concise details about a video's themes and objects, helping the system better understand content and enhance user-content matching (Panniello et al. 2016). In organic recommendations, for instance, content-based

[10] See https://www.multicollab.com/blog/user-generated-content/.

[11] Rather than directly processing raw video streams, which is computationally intensive, Platform A uses extracted features from these streams in its recommender system. These features are generated by a separate algorithm within the platform. This design can provide a more efficient video recommendation process.



(a) Control Group (b) Treatment Group

Figure 2 Using AI-generated Titles on Producers' Video Posting Page

algorithms utilize video titles to recommend videos similar to what users have already watched (Adomavicius et al. 2008). For users who frequently watch videos with titles containing keywords such as "cooking" or "recipes," the system can recommend other videos with related keywords. In search-query recommendations, titles provide direct text matches to user searches, addressing challenges in cross-modality matching by offering clear textual references that improve accuracy. In our study, organic recommendations drove the vast majority of video consumption, and search-oriented recommendations only accounted for less than 1% of total viewership. Titles are also vital in addressing the cold-start problem, where new videos without engagement data rely heavily on descriptive titles to be categorized and recommended (Wei et al. 2024). Additionally, titles help mitigate the effects of noisy engagement data, ensuring stable and relevant recommendations even when user engagement is inconsistent due to short content life cycles (Davidson et al. 2010).

Producers' title generation can generally be described as a two-step process (see Figure C.1). First, producers decide whether to provide a title. Second, when a title is provided, producers decide how much effort to put into creating the title, which can result in differentiated quality. Some create simple titles, whereas others craft richer ones that better capture the video's themes.

3.2. Experiment Design

Like many UGC platforms, Platform A faces the significant challenge of metadata sparsity. In our dataset (see Section 3.3), only 60.7% of videos had titles during the pre-treatment period. To address this, Platform A introduced GAI tools in July 2023, which leveraged a multimodal large language model capable of processing text and images. Platform A fine-tuned this model by curating a manually selected dataset of videos with well-matched titles as the training set. The fine-tuning

process allows the model to learn patterns in the relationship between video content (both visual and textual) and suitable titles on Platform A. To generate metadata for new videos, as shown in Figure C.2, the platform's GAI tool captured multiple frames from the video stream, extracting visual elements (e.g., key objects or scenes) and any text present in the frames (e.g., subtitles, on-screen text). This combination of visual and textual data was then fed into the fine-tuned model, which produced a title that best reflects the input content. When new videos are uploaded, the GAI tool could generate relevant titles automatically. This process allows the platform to generate coherent metadata without requiring human input for every video.

To causally examine the value of AI-generated metadata, we conducted a field experiment on the video posting page to simulate the metadata input change that feeds into the video recommender system. This experiment lasted from July 20th to August 21st, 2023. Producers involved in our experiment were randomly assigned to the control and treatment groups. During the experiment, all videos uploaded by treatment-group producers were supposed to have their title-setting boxes on the video posting page automatically populated with AI-generated titles, along with a notification indicating that the titles were generated by AI (see Figure 2(b)). In contrast, for control group producers, the title-setting box remained blank after upload (see Figure 2(a)). In addition, in 2023, external GAI tools for generating video titles were unlikely to be widely used since major language models did not support video processing, and few video platforms offered such features. Thus, the risk of contamination, where the control group was unintentionally influenced by the experimental intervention, was limited. Treatment producers had the flexibility to either delete, amend, or fully adopt the AI-generated title, and regardless of their choices, viewers could not see any indication in their interface of whether the titles were generated by AI. In addition, as shown in Figure 1, video titles were small and positioned at the bottom left corner of the viewers' interface, so we assumed that any observed changes in video consumption and engagement between the treatment and control groups were unlikely to be driven by changes in the visibility of the titles to viewers[12].

3.3. Data and Variables

Platform A only stored the AI-generated titles between August 8th and August 21st during the experiment due to technical issues. Thus, our dataset was segmented into two periods: (1) pre-treatment period: July 10th to July 19th and (2) treatment period: August 8th to August 21st. Our study included $2,048,033$ producers who posted at least one video during our treatment period, with 1,024,940 in the treatment group and 1,023,093 in the control group. During the treatment period, producers in the treatment group uploaded 5,377,560 videos, while those in the control group posted 5,361,424 videos. Among the 5,377,560 videos uploaded by the treatment group,

[12] We will provide evidence for this assumption in Section 5.2.2.

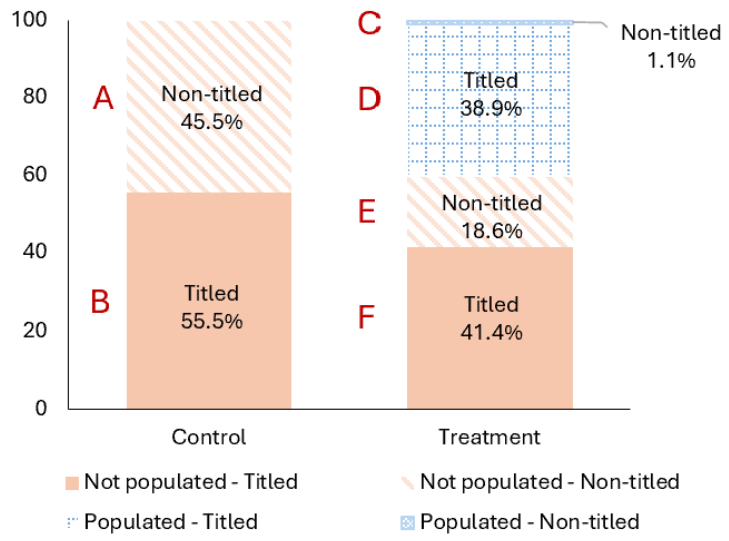


Figure 3 Proportion of Videos With Title-Setting Boxes Populated by AI-Generated Titles

Notes: Panels A-F denote corresponding groups of videos: A = control group, not populated (with AI-generated titles) and non-titled; B = control group, not populated and titled; C = treatment group, populated and non-titled; D = treatment group, populated and titled; E = treatment group, not populated and non-titled; F = treatment group, not populated and titled. AI-generated titles were not populated for some treatment group videos (Panels E and F) due to technical issues (e.g., system delays or timeouts).

3,226,222 (60%) videos did not receive AI-generated titles (see Figure 3). This was primarily due to technical limitations in the AI title generation process. For example, some producers completed posting before the system could return a title, the process occasionally timed out under high load, or the generation failed due to technical issues.[13] During the pre-treatment period, only 60.7% of videos had titles, indicating prevalent metadata sparsity on Platform A. Among the 2,151,338 treatment group videos that received AI-generated titles, 97.25% had titles, highlighting the significant role of AI-generated metadata in alleviating metadata sparsity. For each producer, we obtained data for video viewership outcomes, producer characteristics, and video characteristics. To accommodate variations in video posting times, we calculated the cumulative viewership outcomes for each video over the first two weeks after posting (Zeng et al. 2023).

Table D.1 of Online Appendix D presents a summary of variables used in our analysis. Our independent variable is the treatment group dummy ($Treat_i$), coded as 1 if producer i was assigned to the treatment group. Dependent variables, which capture viewers' video consumption, include viewership metrics such as the number of valid watches ($ValidWatch_{ij}$), and viewers' total watch duration in minutes ($WatchDuration_{ij}$). To analyze the heterogeneous effects of AI-generated

[13] Producer-initiated deletion or replacement of AI-generated titles is recorded as an edit and is not classified as a non-populated case.

titles, we used a moderator variable: $Utilitarian_{ij}$, indicating whether video j of producer i was a utilitarian-content video, coded as 1 for know-how and news categories, and 0 otherwise.[14]

To account for various confounding factors, we include a set of control variables ($Controls_{ij}$) reflecting producer, video, and day attributes as follows. First, we include producers' follower counts and a dummy indicating whether they were key opinion leaders to control producer popularity. Second, we include producers' tenure on the platform (in years), the number of users they followed, and a dummy for multi-homing presence on the focal and rival platforms to control for producers' experience and/or expertise level. Third, we include gender and provincial location dummies to control for producers' demographic and geographic variations. Fourth, we include dummy variables indicating whether the video was publicly visible and whether the video content was composed of actual videos or static images to account for video type. Fifth, we include the video's duration and a binary indicator of whether the producer manually sets a video cover to control for video quality. Sixth, we include dummies for video posting dates and categories to control for temporal and categorical variations. Table D.2 of Online Appendix D presents the summary statistics of our focal variables. A correlation matrix of these variables is shown in Table E.1 of Online Appendix E. To protect Platform A's sensitive information[15], the mean values of four key variables (*ValidWatch*, *WatchDuration*, *Follower*, and *Following*) presented in the tables have been scaled by multiplying random positive numbers.

3.4. Randomization Check

To verify the randomization effectiveness, we compared treatment producers (N=1,024,940) and control producers (N=1,023,093) on their pre-treatment video engagement outcomes, producer characteristics, and video attributes. Pairwise t-tests results in Table F.1 of Online Appendix F show no significant differences between the two groups on these attributes. These results confirm that the treatment and control producers in our sample were comparable, suggesting that any difference between the two conditions after the experiment started should be attributed to our experimental manipulation, i.e., whether producers had access to AI-generated titles.

4. Producer Interactions and Responses

This section addresses RQ1 by identifying producers' interaction patterns with AI-generated titles, examining the effects of AI-generated titles on video production, and analyzing the factors associated with producers' interaction patterns.

[14] Know-how categories generally include educational or instructional videos aimed at elucidating practical skills and knowledge. News categories generally include political news and current affairs. The category classifications are developed by Platform A.

[15] The authors have a Non-Disclosure Agreement (NDA) with Platform A.

4.1. Interaction with AI-Generated Titles

We first investigated how producers interacted with the AI-generated titles using two complementary data sets: (1) survey responses on why and how producers adopt or revise the titles, and (2) paired records of AI-generated and the actual titles adopted by producers. Leveraging the paired title records, we employed an LLM (i.e., GPT-5) to summarize common editing patterns and manually verify them against the survey responses. This process yielded three behavioral patterns (see representative examples in Table 1):

(1) **Rejection (3.2%).** Delete the AI-generated title and leave the field blank.

(2) **Direct adoption (59.5%).** Accept the AI-generated title without modification or with only minor formatting adjustments.

(3) **Human-AI co-creation (37.3%).** Revise the AI-generated titles in one of two ways[16]:

- **Context revision (21.0%).** Revise factual details not evident from the video or not surfaced by the AI output (e.g., place names, product attributes, event context).
- **Emotion revision (16.3%).** Revise adjectives or subjective elements (e.g., personal feelings).

Table 1 Representative Examples of Producer Responses to AI-Generated Titles

Pattern		AI-Generated Title	Actual Title
Rejection		Childhood memories	[Blank]
Direct adoption		Sunflowers blooming in the sunlight	Sunflowers blooming in the sunlight
Human-AI co-creation	Context revision	Miao culture, endless heritage #MiaoCulture	Ethnic culture #Xigu Village, Puxiong Town, Yuexi County #TorchFestival
	Emotion revision	Mountain view	The view from the mountain summit was worth every ounce of effort to scale it

Notes: The examples are drawn from actual cases in our dataset.

4.2. Effects on Video Production

We next examined whether AI-generated titles affected producers' video production behaviors by conducting producer-level t-tests on the number of videos produced and the average time gaps (in hours) between consecutive video uploads for treatment and control groups during the experiment. As shown in Table G.1 in Online Appendix G, we find no significant differences in either measure. These results indicate that the intervention influenced titling decisions rather than video production. This is consistent with Platform A's video posting process, in which titles are generated only after videos have been uploaded.

[16] This aligns with prior information systems (Chen and Chan 2023) and recommender systems literature (Golubovikj and Tkalčič 2025), which conceptualizes content as comprising contextual (i.e., factual or informational) and emotional (i.e., sentiment or subjective) attributes that underpin content understanding and recommendation performance.

4.3. Factors Influencing Interaction Patterns

We next examined producer and video attributes associated with each interaction pattern detailed in Section 4.1. We employed an LLM (i.e., GPT-5) to automatically label observations of paired AI- and human-generated titles. Table 2 summarizes the labeling procedure (Cascio Rizzo et al. 2024). Manual validation shows that the LLM achieved an average accuracy of 90.1%.

Table 2 Process of Labeling Producer Responses to AI-Generated Titles

Step	Description
1	Define the classification scheme and coding rules
2	Train human annotators and label a reference sample
3	Prompt the LLM using guidelines and few-shot examples from Steps 1-2
4	Refine prompts and coding rules based on LLM feedback and errors
5	Test and evaluate LLM performance on a held-out sample
6	Deploy the LLM to code the full dataset
7	Manually validate LLM outputs on a subsample using human raters

Notes: The labeling process follows a human-in-the-loop design to ensure consistency and reliability.

This procedure yielded four mutually exclusive dummy variables: *Reject* (rejection), *Adopt* (direct adoption), *ContextRevise* (context revision), and *EmotionRevise* (emotion revision). We then used these dummies as dependent variables in a series of probit models specified as:

$$\Pr(Outcome1_{ij} = 1) = \Phi\big(\gamma_0 + \gamma_1\, Controls_{ij}\big), \tag{1}$$

where $Outcome1_{ij}$ represents one of the four interaction dummies adopted by producer i for video j. $Controls_{ij}$ include all aforementioned producer-, video-, and day-level attributes. Highly skewed control variables were log-transformed. γ_1 contains the focal coefficients of interest.

Table 3 reports the estimation results of Equation (1). The positive coefficient of *Utilitarian* in Column (2) shows that producers are more likely to directly adopt AI-generated titles for utilitarian-content videos than for hedonic-content ones. Titles for utilitarian-content videos often describe tangible elements of the video stream (e.g., objects, products, or events), whereas titles for hedonic-content videos (e.g., selfies) often convey personal feelings or subjective expression. Because GAI tools generate titles from raw video streams, AI-generated titles may naturally align better with utilitarian content. Moreover, the positive coefficient of *Utilitarian* in Column (3) and the negative one in Column (4) suggest that producers are more inclined to modify contextual details rather than emotional elements for utilitarian-content videos, consistent with the functional and informational nature of such videos.

For producer characteristics, the coefficients of *Follower*, *Experience*, and *Multihome* are negative in Column (2). These results indicate that producers with more followers, greater experience,

Table 3 Model Estimation Results of Factors Influencing Interaction Patterns

Variables	*Reject* (1)	*Adopt* (2)	*ContextRevise* (3)	*EmotionRevise* (4)
Utilitarian	-0.204***	0.030***	0.035***	-0.024***
	(0.010)	(0.006)	(0.006)	(0.006)
Follower	0.023***	-0.095***	0.063***	0.045***
	(0.002)	(0.002)	(0.002)	(0.002)
KOL	-0.355***	-0.275***	0.438***	-0.188***
	(0.065)	(0.046)	(0.035)	(0.031)
Experience	0.031***	-0.010***	0.002	0.006***
	(0.002)	(0.002)	(0.001)	(0.002)
Following	-0.060***	0.158***	-0.121***	-0.059***
	(0.002)	(0.002)	(0.002)	(0.002)
Multihome	0.029***	-0.227***	0.180***	0.129***
	(0.006)	(0.006)	(0.005)	(0.005)
Female	-0.063***	-0.041***	-0.085***	0.179***
	(0.006)	(0.006)	(0.004)	(0.005)
PublicVisible	-0.184***	-0.727***	0.607***	0.562***
	(0.008)	(0.009)	(0.008)	(0.008)
VideoDuration	0.032***	-0.250***	0.173***	0.185***
	(0.007)	(0.008)	(0.006)	(0.006)
Cover	0.084***	-0.778***	0.497***	0.443***
	(0.007)	(0.006)	(0.005)	(0.005)
ContentType	-0.244***	0.901***	-0.607***	-0.483***
	(0.006)	(0.007)	(0.005)	(0.005)
Controls	Yes	Yes	Yes	Yes
Observations	2,151,338	2,151,338	2,151,338	2,151,338
Log-likelihood	-293,817.1	-1,239,444.3	-996,921.1	-903,368.2

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors. Dummies for provincial location, video posting dates, and video categories were included in model estimation but not reported in the table.

and multi-homing behaviors are less likely to adopt them without modification and instead tend to revise or reject them. These results suggest that producers with greater knowledge of title generation, whether through skill, experience, or exposure to multiple platforms, are less likely to entirely adopt AI-generated titles and more inclined to exercise editorial control.

We also observe heterogeneous effects by gender. The coefficient of *Female* in Column (4) is significantly positive, showing that female producers are more likely to add emotion revision to AI-generated titles, consistent with prior evidence that women tend to use more affective elements in communication (Coates 2015). Finally, the coefficients of *PublicVisible* are significantly positive in Columns (3) and (4), suggesting that producers are more likely to revise titles with contextual or emotional elements when videos are publicly visible. When producers anticipate broader audience exposure in public settings, they will invest more effort in refining the titles.

5. Economic Value of AI-Generated Metadata

This section addresses RQ2 by quantifying the impact of AI-generated metadata on video consumption and examining the underlying mechanisms.

5.1. AI-Generated Metadata Boosts Content Consumption

We next examined the effect of treatment (i.e., access to AI-generated titles) on video viewership. Our unit of analysis was at the producer-video level to capture changes in viewership outcomes for each video uploaded by producers. We estimated the following ordinary least squares (OLS) regression with robust standard errors:

$$Outcome2_{ij} = \beta_0 + \beta_1 Treat_i + \beta_2 Controls_{ij} + \epsilon_{ij} \tag{2}$$

where $Treat_i$ is a binary indicator equal to 1 if the producer i was in the treatment group, $Controls_{ij}$ include all aforementioned producer-, video-, and day-level attributes, and ϵ_{ij} is the error term. $Outcome2_{ij}$ represents our two viewership metrics, i.e., $ValidWatch_{ij}$ (number of valid watches for video j of producer i) and $WatchDuration_{ij}$ (viewers' total watch duration). All dependent variables and highly-skewed control variables were log-transformed and incremented by one to accommodate zero values, following the semi-log specification in Cole and Sokolyk (2018).

The model estimation results in Table 4 show that AI-generated titles boosted video viewership. Specifically, Column (1) indicates an increase of 1.6%[17] in valid watches in the treatment group compared to the control group ($\beta_1 = 0.016$, p-value < 0.01). Results in Column (2) indicate that treatment group videos enhanced watch duration by 0.9% from the control group ($\beta_1 = 0.009$, p-value < 0.05).[18] Given our sample covers 2% of all video producers on platform A, who posted over 10 million videos during our experiment, this result translates to billions of additional valid watches and billions of extra minutes in watch duration across the platform, demonstrating significant economic benefits. To contextualize the practical implications of our findings, we conduct a back-of-the-envelop cost-benefit analysis. Platform A handles approximately 35.7 million new video uploads per day. The platform generates titles with A100 GPUs (about 1 GPU second per title). One A100 GPU hour is priced at about $4 by cloud service providers such as AWS, so the total cost to generate titles for all videos is about $40,000 per day. Given the platform's average daily ad revenue of about $276 million, a 1.6% increase in valid watches translates to about $4.4 million in additional daily advertising revenue (assuming stable ad load and cost-per-mille, CPM), which should outweigh the cost of title generation.

[17] The marginal effect size is calculated as: Exp(0.016) - 1 = 1.6%. The same calculation method is applied throughout this paper.

[18] In Appendix H, we further estimate the local average treatment effect (LATE), focusing on treatment videos for which AI-generated titles were successfully populated (i.e., groups C and D in Figure 3). The results remain qualitatively consistent, while the estimated coefficients are larger in magnitude.

Table 4 Results of Having Access to AI-Generated Titles on Content Consumption

Variables	*ValidWatch* (1)	*WatchDuration* (2)
Treat	**0.016*****	**0.009****
	(0.004)	(0.004)
Relative Effect Size	1.6%	0.9%
Controls	Yes	Yes
Observations	10,738,984	10,738,984
R-squared	0.297	0.322

Notes: $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.1$. Values in parentheses are robust standard errors clustered at the producer level.

5.2. AI-Generated Metadata Improves Recommendation Accuracy

So far, we have shown that AI-generated titles significantly increased viewership outcomes. Four mechanisms could explain this effect: (1) improved user-video matching by enabling the recommender system to better understand video content, (2) direct influence on viewers through title visibility in the viewer interface, (3) algorithmic promotion if the platform boosted exposure of videos with AI-generated titles, and (4) viewer-side cannibalization if treatment videos crowded out viewers' engagement with control videos. Below, we tested each mechanism and confirmed that improved user-video matching accuracy of the recommender system drives the treatment effects.

5.2.1 User-Video Matching of Recommender System

We hypothesized that AI-generated titles improved video viewership by enhancing user-video matching accuracy. To illustrate, AI-generated titles could help recommender systems better understand and interpret video content. This enhanced understanding should enable the system to more accurately predict which users are more likely to engage (e.g., view/like/share videos or follow the producer) and match the video with these specific users, thereby improving the video's discoverability and increasing the likelihood of user engagement. This improved user-video matching accuracy translates into the higher viewership outcomes observed in Section 5.1. We tested this hypothesis through two complementary analyses.

Recommendation Accuracy. To directly evaluate user-video matching accuracy, we used the Area Under the ROC Curve (AUC), a widely applied metric in recommender system studies (Chen et al. 2024, Bi et al. 2024). AUC measures how well the model predicts viewer engagement behaviors by comparing the predicted and actual viewer engagement outcomes. A higher AUC indicates more accurate user-video matching. To calculate AUC, we collected a proprietary dataset from the platform's recommender system, documenting video recommendations from November 1st to November 30th, 2023 for videos posted during our experiment.[19] It included 93,618,096 records with

[19] Videos produced during our experiment period were still recommended to users with their titles unchanged after the experiment. Because the platform did not archive the predicted engagement data during our experiment period, we used the data collected in November, 2023.

details on predicted engagement probabilities (i.e., like videos, share videos, and follow producers) and actual viewer behaviors for each user-video pair.

While our main analysis focuses on viewership outcomes (e.g., valid watch and watch duration), this additional dataset does not include predictions for these measures. Instead, it focuses on downstream engagement behaviors, which occur at later stages of the user journey. These predicted behaviors serve as essential inputs for the recommender system to match users with content that aligns with their preferences. Higher AUC values for like, share, and follow indicate that the recommender system effectively predicts user interactions. As these behaviors occur at later stages of the user journey, their accurate predictions imply that viewership outcomes, which occur earlier, are also likely to be predicted accurately. Collectively, a higher AUC for these engagement behaviors reflects improved user-video matching accuracy and supports our hypothesis that AI-generated titles enhance video viewership and engagement through better user-video matching.

To compare whether AUC values differ significantly across treatment and control group videos, we performed 1,000 bootstrap resampling iterations to calculate AUC and p-values for both treatment and control videos. The results in Table 5 showed significant improvements. The AUC for shares increased from 0.823 in the control group to 0.848 in the treatment group, an improvement of 0.026 (p-value$<$0.01). For likes, the AUC rose from 0.892 to 0.921, an increase of 0.029 (p-value$<$0.01), and for follows, it increased from 0.867 to 0.887, a rise of 0.019 (p-value$<$0.01). These results confirmed that AI-generated titles significantly improved user-video matching accuracy. These findings support our hypothesis that AI-generated titles enhance video viewership and engagement through better user-video matching.

Table 5 AUC Comparison

Variables	Treatment Group	Control Group	Difference	P-value
Share	0.848	0.823	0.026	**<0.01**
Like	0.921	0.892	0.029	**<0.01**
Follow	0.887	0.867	0.019	**<0.01**

Recommendation Channels. To complement the system-level evidence on AUC difference, we next examined whether the effect of AI-generated titles held consistently across Platform A's two main recommendation channels, i.e., organic and search-oriented recommendations (see Section 3.1). Using detailed consumption data, we identified the source channel of each video watch, aggregated valid watch counts and watch duration by channel, and re-estimated Equation (2) by channel. The results in Table 6 show a 0.3% increase in both valid watches and watch duration for treatment videos in the search-oriented channel, and stronger gains of 1.5% and 0.9% in the organic channel. These findings highlight that AI-generated titles consistently

enhanced content consumption regardless of recommendation channels, reinforcing the mechanism of improved user-video matching. The larger boost in the organic channel likely occurred because search-oriented recommendations already involved user input, leaving less room for improvement in matching.

Table 6 Results for Search-Oriented and Organic Recommendations

	Search-Oriented Recommendation		Organic Recommendation	
Variables	*ValidWatch*	*WatchDuration*	*ValidWatch*	*WatchDuration*
	(1)	(2)	(3)	(4)
Treat	**0.003*****	**0.003****	**0.015*****	**0.009****
	(0.001)	(0.001)	(0.004)	(0.004)
Controls	Yes	Yes	Yes	Yes
Observations	10,738,984	10,738,984	10,738,984	10,738,984
R-squared	0.166	0.165	0.295	0.321

Notes: $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.1$. Values in parentheses are robust standard errors clustered at the producer level.

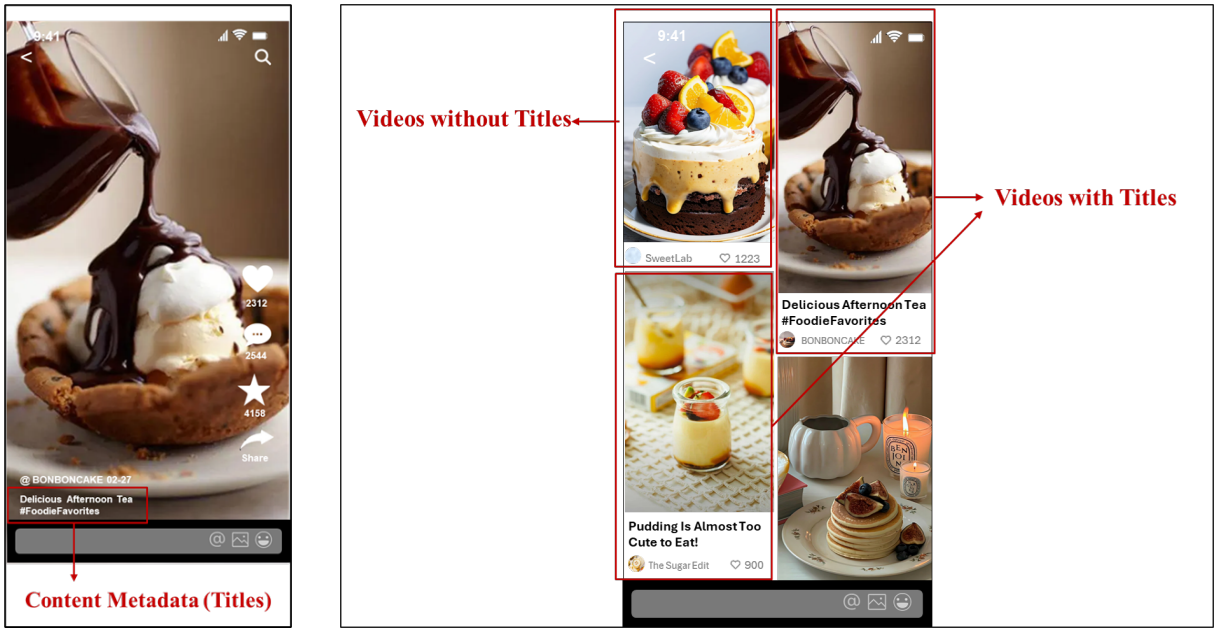
5.2.2 Title Visibility

As shown in Figure 1, titles were displayed in the bottom-left corner of the viewer interface, less prominent than other visual elements. However, titles might still attract viewer attention and affect video consumption outcomes. We therefore tested the role of title visibility through two analyses.

AI-Generated Titles under Different User Interfaces (UIs). To examine whether video titles directly drive consumption outcomes, we compared two user interfaces on Platform A: single-feed and dual-feed. In the single-feed UI (Figure 4(a)), users scrolled vertically through auto-playing videos, and titles vaguely appeared in the lower-left corner over the video background, making them relatively less prominent. In contrast, the dual-feed UI (Figure 4(b)) displayed two vertical columns of non-auto-playing videos.[20] Titles were shown below each video cover against a white background, making them more visually prominent to viewers than the single-feed UI (see Figure 4(b)). If title visibility directly affects viewer behavior, e.g., by helping viewers better assess a video's relevance or anticipate its content, then the treatment effect should be stronger under the dual-feed UI, where titles are more visually prominent. To test this hypothesis, we estimated Equation (2) separately for single- and dual-feed UIs. Using the delta method, we find no significant differences in relative effect sizes across UIs (see Table 7), suggesting that the observed boost in viewership is not primarily driven by direct visual exposure to titles.

Diminishing Treatment Effects with Accumulated Video Consumption. Beyond the UI comparison, we tested title visibility by examining whether the treatment effects diminished as

[20] Dual-feed format accounts for about 25% of the total consumption on Platform A.



(a) Single-feed interface (b) Dual-feed interface

Figure 4 Single/Dual-feed interface: Titles appear (a) faintly within videos and (b) prominently below video covers

Table 7 Comparison of Treatment Effects Between Single- and Dual-Feed Interfaces

Variables	*ValidWatch*		*WatchDuration*	
	Single-feed	Dual-feed	Single-feed	Dual-feed
	(1)	(2)	(3)	(4)
Treat	**0.013*****	**0.020*****	**0.011*****	**0.013****
	(0.004)	(0.004)	(0.003)	(0.003)
Controls	Yes	Yes	Yes	Yes
Observations	10,738,984	10,738,984	10,738,984	10,738,984
R-squared	0.233	0.252	0.262	0.264
Delta Method p-value	**0.769**		**0.749**	

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors clustered at the producer level.

videos accumulated more valid watches. Each additional valid watch provides the platform with another observed viewer-video interaction data point, which helps the recommender system learn which users are likely to engage with the video. If title visibility was the main driver, the treatment effect should remain relatively stable as a video accumulates more interactions, because the visibility of the title to viewers does not depend on how much the system has already learned about the video. By contrast, if improved user-video matching was the main mechanism, the marginal benefit of metadata should decline over time, because recommender systems can increasingly rely on accumulated interaction data rather than metadata to infer which users are the best match for the video. Using videos posted on August 8th, 2023, and tracking them for 14 days (as detailed in

Online Appendix I), we find that the treatment effect weakened as videos accumulated more valid watches, indicating that user-video matching is likely to be the primary mechanism.

5.2.3 Algorithmic Promotion

We next examined algorithmic promotion as another potential mechanism. If Platform A had intentionally boosted exposure for treatment videos to encourage GAI adoption, these videos would receive more views, naturally increasing viewership outcomes even without better matching. To evaluate this, we normalized key outcomes (i.e., valid watches, watch duration) by total video views to hold exposure constant and re-estimated Equation (2). The treatment effects remain significant (detailed in Online Appendix J), and platform engineers confirmed that no additional prioritization occurred, indicating that algorithmic promotion does not explain the results.

5.2.4 Viewer-Side Cannibalization

We next tested viewer-side cannibalization by examining whether increased consumption of treatment videos crowded out viewers' engagement with control videos. Using a user-day-level dataset of 10,000 randomly sampled viewers, we regressed viewers' valid watches of control videos on the share of treatment videos they watched. We found that the share of treatment videos watched was not negatively associated with valid watches of control videos (detailed in Online Appendix K), indicating that the treatment effect is not driven by viewer-side cannibalization.

6. How Should Platforms Best Leverage AI-Generated Metadata?

This section addresses RQ3 by examining how AI-generated titles affect the title generation process, to inform how platforms can best leverage AI-generated metadata. As producers' title generation follows a two-step process (Section 3.1), the impact of AI-generated titles can be characterized along two dimensions. The first is a coverage effect, which captures whether AI reduces metadata sparsity by filling in titles for videos that would otherwise lack them. The second is a quality effect, which arises conditional on a title being present and reflects whether AI enhances the richness and descriptiveness of titles. We structure the following analyses around these two effects.

6.1. Coverage Effect

We first examined the textual characteristics of video titles using alternative dependent variables in Equation (2). We utilized two binary indicators: $HasTitle_{ij}$, which indicates whether video j had a title (either GAI or human-generated), and $HasTag_{ij}$, which denotes whether the video title included tags. The analysis uses the full sample of treatment videos (i.e., Panels C+D+E+F in Figure 3) and control videos (i.e., Panels A+B in Figure 3). After re-estimating Equation (2) with these variables, results in Table 8 demonstrate that having access to AI-generated titles increased

the likelihood of a video having a title by 41.4% and enhanced the probability of having tags by 72.4%[21]. These results suggest that having access to AI-generated titles effectively reduced metadata sparsity by increasing title and tag completeness, which in turn boosted viewership.

Table 8 Results of Video Title Characteristics Analysis

Variables	*HasTitle* (1)	*HasTag* (2)
Treat	**0.244*****	**0.247*****
	(0.001)	(0.001)
Control Baseline (Mean)	0.590	0.341
Relative Effect Size	41.4%	72.4%
Controls	Yes	Yes
Observations	10,738,984	10,738,984
R-squared	0.165	0.126

Notes: $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.1$. Values in parentheses are robust standard errors clustered at the producer level.

Building on these findings, we next examined whether this viewership-boosting effect was stronger for videos more affected by metadata sparsity. We introduced a moderator variable based on content type, as informed by prior research. According to social exchange theory, digital content creators are driven by motives like personal fulfillment or follower growth for financial gain (Wasko and Faraj 2005). Utilitarian-content videos (e.g., news and reviews) tend to have more detailed metadata to attract followers, while hedonic-content videos (e.g., personal vlogs) typically lack such detail due to being more focused on self-expression. Accordingly, we constructed a moderator variable: $Utilitarian_{ij}$, identifying whether a video was of a utilitarian type.

The theoretical prediction above is confirmed by our data. During the pre-treatment period, 61.3% of utilitarian-content videos had titles, compared to 57.3% of hedonic-content videos. Therefore, we hypothesize that AI-generated metadata may have a negative moderation effect for utilitarian-content videos. To test this hypothesis, we incorporated the moderator variable and its interaction terms with $Treat_i$ in Equation (2).

Results in Table 9 show that utilitarian-content videos in the treatment group experienced a relative decrease in valid watches by 3.1% (p-value<0.05) and watch duration by 3.0% (p-value<0.05) than hedonic-content videos. Overall, utilitarian-content videos in the treatment group showed a 1.4%[22] decrease in valid watches, and a 1.9% decrease in watch duration, which is likely due to the more detailed human-generated metadata already associated with utilitarian-content videos. Altogether, the viewership-boosting effects of AI-generated metadata are weaker (stronger) for utilitarian-content (hedonic-content) videos.

[21] The relative effect size is calculated as 0.244/0.590=0.414 and 0.247/0.341=0.724.

[22] This is calculated as: Exp(-0.032+0.018)-1 = -1.4%.

Table 9 Heterogeneous Effects of AI-Generated Metadata Access on Content Consumption Across Video Types

Variables	*ValidWatch* (1)	*WatchDuration* (2)
Treat	0.018***	0.011***
	(0.004)	(0.001)
*Treat * Utilitarian*	**-0.032****	**-0.030****
	(0.013)	(0.012)
Controls	Yes	Yes
Observations	10,738,984	10,738,984
R-squared	0.297	0.322

Notes: ***p<0.01, **p<0.05, *p<0.1. Values in parentheses are robust standard errors clustered at the producer level. *Utilitarian* is absorbed by the video category dummies in *Controls* and therefore not reported in the table.

6.2. Quality Effect

We next turn to the quality effect. Conceptually, title quality refers to how well a title conveys video-relevant information in a way that helps recommender systems and viewers understand the video once a title is present. Accordingly, unlike the coverage effect, which concerns whether a title is provided at all, the quality effect is evaluated by how viewership differs across human-written, AI-generated, and human-AI co-created titles conditional on title presence. To examine this quality effect, we proceeded in four steps. First, we compared AI-generated and human-written titles. Second, we examined whether human-AI co-created titles outperformed AI-generated and human-written titles in viewership. Third, we compared these title types by lexical richness as an indicator of content quality. Finally, we examined the effects of context revision and emotion revision on viewership and lexical richness.

6.2.1 AI-Generated vs. Human-Generated Titles' Quality

To isolate this quality effect, we restricted the analysis to titled videos and further excluded 2,226,922 treatment videos that did not actually receive AI-generated titles due to algorithmic delivery issues (see details in Section 3.3). The analysis therefore focused on titled videos in the control group (i.e., Panel B in Figure 3) and titled videos with AI-generated titles populated (i.e., Panel D in Figure 3). However, a direct comparison between titled videos in the treatment and control groups would be misleading. This is because some videos in the treatment group may only be titled because of the access to AI-generated titles, potentially indicating lower producer effort and video quality. Thus, such a direct comparison could underestimate the true effect of accessing AI-generated titles on viewership for titled videos. To address this, we used the propensity score matching (PSM) method with the radius matching algorithm[23], employing one-to-many matching

[23] We also applied other matching algorithms (e.g., kernel matching) and found robust results.

to pair each titled video in the treatment group with several of the most "similar" titled videos in the control group based on pre-treatment covariates (more details are available in Appendix L). In particular, we included producer-level variables (e.g., experience, prior activity) that proxy for producer effort and knowledge, as well as video-level variables (e.g., length, category, upload time) and historical performance indicators (e.g., prior views and engagement) that proxy for video quality and audience appeal. Conditional on these covariates, treatment assignment for titled videos can be regarded as quasi-random, thereby satisfying the conditional independence assumption underlying the PSM approach. Using the matched sample, we re-estimated Equation (2), applying weights to account for the multiple matches per treated unit. Interestingly, the results in Table 10 show that videos in the treatment group experienced a decrease of 37.9%[24] in valid watches and 32.6% in watch duration compared to the control group. These results suggest that AI-generated titles may not outperform existing human-generated titles in terms of quality.

Table 10 Model Estimation Results for Titled Videos (Matched Sample)

Variables	*ValidWatch* (1)	*WatchDuration* (2)
Treat	**-0.476*****	**-0.394*****
	(0.002)	(0.001)
Relative Effect Size	-37.9%	-32.6%
Controls	Yes	Yes
Observations	3,885,089	3,885,089
R-squared	0.329	0.351

Notes: $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$. Values in parentheses are robust standard errors. The control baseline used here is calculated based on the matched sample.

6.2.2 Human-AI Co-Created Titles' Quality

To further explore this phenomenon, we examined how the extent of human revision to AI-generated titles shaped viewership outcomes. Following prior text-mining research (Burtch et al. 2022), we constructed a measure of textual similarity using cosine similarity between TF-IDF representations of text.[25] Specifically, we computed the cosine similarity between the original AI-generated title and the final title adopted by the producer ($Similarity_{ij}$), using a broader corpus of video titles

[24] The marginal effect is calculated as: Exp(-0.476)-1 = -37.9%. The same calculation method is applied subsequently.

[25] TF-IDF is a scaled matrix that captures how frequently each term appears in a document relative to its frequency across all documents in the dataset. This scaling decreases the weights of common words that are less informative for distinguishing between documents, thus emphasizing words that are unique to specific documents. If a word is highly unique and only appears in a single document, its impact is preserved, while words common across many documents are given less weight. Calculating cosine similarity between document vectors in this TF-IDF-derived space effectively measures similarity in terms of distinctive word usage.

from both the treatment and control groups. Lower similarity indicates more extensive human revision of the AI-generated title, whereas higher similarity indicates that the producer adopted the AI-generated title with fewer changes. Additional validation and interpretation of this similarity measure are reported in Online Appendix M.

We then incorporated this similarity measure ($Similarity_{ij}$) into Equation (2) by adding an interaction term between $Treat_i$ and $Similarity_{ij}$:[26]

$$Outcome2_{ij} = \alpha_0 + \alpha_1 Treat_i + \alpha_2 (Treat_i \times Similarity_{ij}) + \alpha_3 Controls_{ij} + \mu_{ij} \tag{3}$$

In this specification, the coefficient on $Treat_i$ captures the treatment effect when $Similarity_{ij} = 0$, that is, when the final producer-adopted titles have almost no overlap with the original AI-generated titles. As shown in Table 11, the positive coefficient of $Treat_i$ indicates that when producers substantially revised the AI-generated titles, treatment group videos performed better than the comparable control group videos. The coefficient on the interaction term then captures how this treatment effect changes as the final producer-adopted titles become more similar to the original AI-generated titles. Its negative sign indicates that greater similarity between the original AI-generated titles and the final producer-adopted titles was associated with worse viewership outcomes. Specifically, a 10% increase in similarity reduced valid watches by 9.8%[27] and watch duration by 8.2%.

Taken together, these coefficients imply that the benefit of treatment declined as producers made fewer changes to the AI-generated titles. Based on the estimated coefficients, treatment group videos outperformed control group videos only when similarity fell below 20.8%[28] for valid watches and 20.9% for watch duration. These results suggest that adopting AI-generated titles with minimal revision tended to reduce viewership, whereas substantial human revision improved title quality and could reverse the negative effect. These findings suggest that while AI-generated titles reduced production costs and addressed title sparsity, producers should revise these titles rather than adopt them without changes. This pattern emphasizes the benefits of combining human judgment with AI efficiency (Boyacı et al. 2024, Chen and Chan 2023, Zhou and Lee 2024).

To address the concern that low title similarity may simply capture higher producer effort or video quality, we applied inverse probability weighting (IPW) and propensity score matching (PSM). Results based on the weighted and matched samples continue to show higher viewership for

[26] The main effect of $Similarity_{ij}$ is not included in this specification because it would be absorbed by the interaction term ($Treat_i \times Similarity_{ij}$) due to collinearity.

[27] The marginal effect size is calculated as: $\exp((-1.026) \times 0.1) - 1 = -9.8\%$.

[28] This threshold is calculated as $0.213/1.026 = 20.8\%$. The same method is applied subsequently.

Table 11 Heterogeneous Estimation Results for Titled Videos (Matched Sample)

Variables	*ValidWatch* (1)	*WatchDuration* (2)
Treat	**0.213*****	**0.178*****
	(0.002)	(0.002)
*Treat * Similarity*	**-1.026*****	**-0.852*****
	(0.003)	(0.002)
Controls	Yes	Yes
Observations	3,885,089	3,885,089
R-squared	0.382	0.393

Notes: $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$. Values in parentheses are robust standard errors.

the treatment group (see Online Appendix N). A further concern is that if lower similarity leads to better video viewership, the optimal outcome should occur when similarity equals zero (i.e., no AI contribution). However, zero similarity does not necessarily imply the absence of AI influence. Even when producers did not directly adopt AI-generated titles, they were still exposed to AI suggestions that inspired their own title creation, which we refer to as the *inspiration effect*. To further validate this interpretation, we collected open-ended responses from 1,925 treatment producers in August on how AI-generated titles influenced their title-generation process. Their qualitative feedback consistently highlighted an *inspiration effect*, where users perceived AI-generated titles as creative catalysts[29]. For example, one user noted, "*It's already great; it may not always be precise, but it provides some inspiration for our posts!*" Another mentioned its help with "writer's block," saying, "*When I can't think of anything, it helps a bit.*" These insights show that AI-generated titles act as catalysts in content creation,[30] motivating producers to rewrite or come up with their own titles. These findings align with recent studies on GAI and creativity (Zhou and Lee 2024).

6.2.3 Human-AI Quality Improvement by Lexical Richness

To better understand how increased human input boosted viewership, we used lexical richness, a key linguistic concept in language studies that signals information quality, as our alternative dependent variable. We followed prior research (Qiao et al. 2020) and measured it through multiple dimensions, including lexical density ($LexicalDensity_{ij}$), lexical variation ($LexicalVariation_{ij}$), and entropy ($Entropy_{ij}$). Lexical richness proxies cognitive effort in text crafting and information quality.[31] Results in Table 12 show that each 10% increase in similarity score was associated with

[29] Due to the NDA between the platform and the authors, we were not permitted to perform text mining on the raw survey responses.

[30] Similar arguments can be found in `https://www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/how-generative-ai-can-boost-consumer-marketing`.

[31] For example, Goes et al. (2014) used lexical density to measure the informational value of reviews. Lexical density is the proportion of content words (such as nouns, verbs, and adjectives) to the total number of words in a text.

a 4.8%[32], 3.7%, and 1.1% decrease in lexical density, lexical variation, and entropy, respectively. These results show that titles with lower similarity tend to be more information-rich. Such titles may provide clearer context, relevant keywords, and expressive signals which help the recommender system better interpret video content and improve user-video matching accuracy.[33]

Table 12 Results of Lexical Richness Analysis (Matched Sample)

Variables	*LexicalDensity* (1)	*LexicalVariation* (2)	*Entropy* (3)
Treat	**0.016*****	**0.017*****	**0.011*****
	(0.0002)	(0.0002)	(0.0007)
*Treat * Similarity*	**-0.245*****	**-0.301*****	**-0.442*****
	(0.0002)	(0.0002)	(0.0007)
Control Baseline (Mean)	0.509	0.812	3.889
Controls	Yes	Yes	Yes
Observations	3,885,089	3,885,089	3,885,089
R-squared	0.621	0.674	0.803

Notes: $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$. Values in parentheses are robust standard errors.

6.2.4 Human-AI Quality Improvement by Context and Emotion Revisions

We next examined how human-AI co-creation shaped viewership outcomes. As described in Section 4.1, human-AI co-creation generally takes two forms: *context revision* and *emotion revision* (Chen and Chan 2023, Golubovikj and Tkalčič 2025), where producers revise either factual details (e.g., place names, product attributes) or personal expressions (e.g., experiences, emotions). Such information is often missing from AI-generated titles because the video's raw stream or on-screen subtitles may not explicitly convey it, making it difficult for the GAI tool to infer without human revision. For example, in Table 1, the AI-generated title, "Miao culture, endless heritage #MiaoCulture," was revised to "Ethnic culture #Xigu Village, Puxiong Town, Yuexi County #TorchFestival," adding specific local details and a cultural festival name. These details provide contextual signals that help the recommender system surface the video to users interested in regional culture or traditional events. Similarly, the title "Mountain view" was rewritten as "The view from the mountain summit was worth every ounce of effort to scale it," introducing an

Lexical variation is the ratio of unique words to the total number of words. Entropy quantifies text unpredictability, computed as:

$$Entropy = -\sum_{k=1}^{n} P_k \log P_k \tag{4}$$

where P_k is the probability of each unique word. We also included sentence length as an additional confounder in this analysis.

[32] The marginal effect size is calculated as: (-0.245 * 0.1) / 0.509 = -4.8%.

[33] See https://hivo.co/blog/creating-descriptive-titles-for-content-with-ai-a-how-to-guide.

emotional tone that may resonate with users seeking inspirational content. Both types of revision offer additional semantic or affective cues that improve the system's ability to interpret and recommend the video content (Panniello et al. 2016). To quantify how these editing behaviors affect video viewership, we estimated the OLS model with robust standard errors:

$$Outcome2_{ij} = \alpha_0 + \alpha_1\, ContextRevise_{ij} + \alpha_2\, EmotionRevise_{ij} + \alpha_3\, Adopt_{ij} + \alpha_4\, Controls_{ij} + \varepsilon_{ij} \quad (5)$$

where $Outcome2_{ij}$ represents the viewership outcomes for video j of producer i. $ContextRevise_{ij}$ and $EmotionRevise_{ij}$ are binary variables denoting whether the producer i revised the AI-generated title by modifying its contextual content or emotional expressions for video j. $Adopt_{ij}$ is a binary indicator equal to 1 if the producer directly adopted the AI-generated title without modification or with only minor formatting adjustments and otherwise 0.[34] $Controls_{ij}$ includes all aforementioned control variables, and ε_{ij} denotes the error term.

Table 13 Results of Context Revision and Emotion Revision on Content Consumption

Variables	*ValidWatch* (1)	*WatchDuration* (2)
ContextRevise	**0.202*****	**0.186*****
	(0.012)	(0.011)
EmotionRevise	**0.193*****	**0.197*****
	(0.012)	(0.011)
Adopt	**-0.684*****	**-0.541*****
	(0.013)	(0.012)
Controls	Yes	Yes
Observations	3,885,089	3,885,089
R-squared	0.379	0.391

Notes: $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$. Values in parentheses are robust standard errors.

The estimation results in Table 13 suggest the significant impact of human-AI co-creation. Specifically, the coefficients of *ContextRevise* in Columns (1) and (2) indicate that compared with control group videos, enriching AI-generated titles with contextual details was associated with a 22.4% increase in valid watches and a 20.4% increase in watch duration.[35] Similarly, the coefficients of *EmotionRevise* suggest that adding emotional expressions was associated with a 21.3% increase in valid watches and a 21.8% increase in watch duration.[36] In contrast, the coefficients of *Adopt*

[34] Since the analysis was based on titled videos after matching, observations where producers deleted the AI-generated titles and left the title-setting box blank, were not included. Minor edits refer to very small textual adjustments, such as changes in capitalization, spacing, punctuation, or adding simple markers (e.g., tagging another user with "@") that do not alter the substantive meaning of the title.

[35] These are calculated as: Exp(0.202) - 1 = 0.224 and Exp(0.186) - 1 = 0.204.

[36] These are calculated as: Exp(0.193) - 1 = 0.213 and Exp(0.197) - 1 = 0.218.

imply that directly adopting AI-generated titles was associated with a 49.4% reduction in valid watches and a 41.9% reduction in watch duration relative to the control group.[37]

We next changed the dependent variables in Equation (5) to different measures of lexical richness. Results in Table 14 show that, consistent with the viewership results, both human-AI co-creation modes are associated with significantly higher lexical richness. Specifically, *ContextRevise* is associated with increases of 16.7%[38], 8.5%, and 9.7% in lexical density, lexical variation, and entropy, respectively, while *EmotionRevise* yields increases of 14.9%, 8.9%, and 13.3%. In contrast, directly adopting AI-generated titles without human revision is associated with decreases of 26.5% in lexical density, 24.5% in lexical variation, and 0.2% in entropy. Taken together, these findings indicate that human input may play a critical role in enriching AI-generated titles, which in turn translates into higher video viewership. The results are also robust to an alternative DML-based identification strategy addressing potential endogeneity in producers' revision decisions[39].

Table 14 Results of Context Revision and Emotion Revision on Lexical Richness

Variables	*LexicalDensity* (1)	*LexicalVariation* (2)	*Entropy* (3)
ContextRevise	**0.085*****	**0.069*****	**0.379*****
	(0.001)	(0.001)	(0.004)
EmotionRevise	**0.076*****	**0.072*****	**0.517*****
	(0.001)	(0.001)	(0.004)
Adopt	**-0.135*****	**-0.199*****	**-0.009****
	(0.001)	(0.001)	(0.004)
Control Baseline (Mean)	0.509	0.812	3.889
Controls	Yes	Yes	Yes
Observations	3,885,089	3,885,089	3,885,089
R-squared	0.610	0.668	0.819

Notes: $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$. Values in parentheses are robust standard errors.

7. Additional Analyses and Robustness Tests

We conducted several additional analyses to validate the robustness and generalizability of our findings. First, the cross-group contamination analyses in Online Appendix P.1 examine whether private messaging between producers created spillovers across treatment and control groups, which would raise concerns about SUTVA. For example, treatment-group producers may have informed

[37] These are calculated as: Exp(-0.684) - 1 = -0.494 and Exp(-0.541) - 1 = -0.419.

[38] This is calculated as: 0.085/0.509 = 16.7%. Similar calculation methods are applied subsequently.

[39] A potential endogeneity concern is that producers who revised AI-generated titles may differ systematically from those who did not in unobserved ways, such as motivation or ability, which could bias the estimated effects. To address this concern, we implemented a DML framework using a 1% subsample trained with gradient boosting and random forest models. The re-estimation results remain qualitatively consistent with the main findings (see Online Appendix O).

control-group producers about the availability of AI-generated titles, thereby changing the latter's behavior and contaminating the estimated treatment effect. The results show no evidence of such spillovers. Second, the long-term effects analysis in Online Appendix P.2, using user-month data from August 2023 to August 2024, confirms that the positive impact of AI-generated titles persists over time. Third, the viewership diversity analysis in Online Appendix P.3 demonstrates that treatment group videos have significantly lower HHI, indicating greater diversity. Finally, analyses using alternative operationalizations of key variables (e.g., different viewership outcomes) in Online Appendix P.4 yield consistent results, confirming the robustness of our findings.

8. Conclusions and Discussions

Our study advances existing literature by studying the value of AI-generated metadata, which is distinct from user-facing content, via a randomized field experiment in which AI-generated titles were minimally visible. We document three types of producer response (i.e., rejection, adoption, and human-AI co-creation) and show that access to AI-generated titles changed producers' titling behaviors and increased video viewership, primarily by improving user-video matching. We further separate a coverage effect (i.e., alleviating metadata sparsity) from a quality effect, showing why human revisions, through context and emotion revisions in particular that heighten linguistic richness, outperformed either AI- or human-generated metadata.

Our results can be generalized beyond Platform A and remain meaningful as future GAI tools evolve. Platform A's recommender system is among the most advanced in the industry, yet we still observe sizable viewership gains, suggesting that less sophisticated platforms could benefit even more. Moreover, our finding that human-AI co-creation outperforms both fully AI- and human-generated titles remains relevant as GAI becomes more capable. Although the share of human input in title generation may decline, it remains essential for adding contextual and emotional details that GAI cannot fully capture from raw video streams.

8.1. Practical Implications

Our results shed light on several important managerial implications. First, our study demonstrates that AI-generated metadata can significantly boost video viewership by improving user-content matching. Therefore, we encourage platform owners to invest in GAI tools that generate metadata, which can address operational challenges related to sparse metadata. While much of the focus has been on GAI's ability to create user-facing content (e.g., advertisements and articles), our results emphasize the equally crucial role of AI-generated metadata in improving platform operations and boosting content discovery. This is relevant for platforms where content consumption is primarily driven by recommendations, such as UGC platforms and e-commerce sites.

Second, our findings show that although AI-generated titles may be of lower quality than human-written ones, their presence still leads to higher viewership than having no metadata. This effect is especially pronounced for hedonic-content videos, which tend to suffer more from metadata sparsity. These results suggest that platforms should minimize missing metadata by automatically filling in AI-generated titles or tags in the back-end of recommender systems when producers leave title fields blank or reject the AI-generated metadata.

Third, our findings emphasize the importance of maintaining human-AI co-creation rather than full automation. While AI tools streamline title generation, human revisions significantly enhance metadata quality and engagement outcomes. Instead of automatically integrating AI-generated titles, platforms should display them as editable drafts and encourage producers to refine them. For example, platforms could place prominent reminders or offer traffic awards, to encourage producers to revise AI-generated titles. This can allow producers to inject their creativity and domain expertise to improve metadata.

Fourth, our results show that improved titles with greater linguistic richness are associated with higher viewership. Our producer response analyses further reveal two primary ways producers co-create with AI-generated titles, i.e., context revision and emotion revision, both of which effectively enhance title quality and increase viewership. These findings provide practical guidance for platform design and producer education. For example, platforms can incorporate AI-assisted prompts that nudge producers toward these empirically effective editing strategies (e.g., "add more context" or "express the video's emotion") and develop tutorials or in-product micro-learning modules around these dimensions. In addition, platforms should consider offering workshops or tutorials to equip content producers with the skills needed to effectively use AI tools.

8.2. Limitations and Future Research

Limitations suggest avenues for future work. First, we study content-based metadata generated from videos alone. Future research could test user- and producer-related metadata (e.g., leveraging creator history or audience patterns), and examine how these metadata types interact to optimize matching and consumption. Second, although our design minimizes title-visibility confounds, experiments that use metadata exclusively for recommendation (without displaying metadata to viewers) could offer complementary identification.

References

Adomavicius, Gediminas, Zan Huang, Alexander Tuzhilin. 2008. Personalization and recommender systems. *State-of-the-Art Decision-Making Tools in the Information-Intensive Age*. INFORMS, 55–107.

Agrawal, Saurabh, John Trenkle, Jaya Kawale. 2023. Beyond labels: Leveraging deep learning and llms for content metadata. *Proceedings of the 17th ACM Conference on Recommender Systems*. 1–1.

Ahrens, Achim, Christian B Hansen, Mark E Schaffer, Thomas Wiemann. 2024. ddml: Double/debiased machine learning in stata. *The Stata Journal* **24**(1) 3–45.

Anthony, Callen, Beth A. Bechky, Anne-Laure Fayard. 2023. "collaborating" with ai: Taking a system view to explore the future of work. *Organization Science* **34**(5) 1672–1694.

Bai, Bing, Tat Chan, Dennis Zhang, Fuqiang Zhang, Yujie Chen, Haoyuan Hu. 2022a. The value of logistic flexibility in e-commerce. *Available at SSRN 4206229* .

Bai, Bing, Hengchen Dai, Dennis J Zhang, Fuqiang Zhang, Haoyuan Hu. 2022b. The impacts of algorithmic work assignment on fairness perceptions and productivity: Evidence from field experiments. *Manufacturing & Service Operations Management* **24**(6) 3060–3078.

Banerjee, Akshina, Oleg Urminsky. 2025. The language that drives engagement: A systematic large-scale analysis of headline experiments. *Marketing Science* **44**(3) 566–592.

Becker, Joel, Nate Rush, Elizabeth Barnes, David Rein. 2025. Measuring the impact of early-2025 ai on experienced open-source developer productivity. *arXiv preprint arXiv:2507.09089* .

Berger, Jonah, Katherine L Milkman. 2012. What makes online content viral? *Journal of Marketing Research* **49**(2) 192–205.

Berger, Jonah, Wendy W Moe, David A Schweidel. 2023. What holds attention? linguistic drivers of engagement. *Journal of Marketing* **87**(5) 793–809.

Bi, Xuan, Mochen Yang, Gediminas Adomavicius. 2024. Consumer acquisition for recommender systems: A theoretical framework and empirical evaluations. *Information Systems Research* **35**(1) 339–362.

Boyacı, Tamer, Caner Canyakmaz, Francis de Véricourt. 2024. Human and machine: The impact of machine input on decision making under cognitive limitations. *Management Science* **70**(2) 1258–1275.

Buell, Ryan W, Tami Kim, Chia-Jung Tsay. 2017. Creating reciprocal value through operational transparency. *Management Science* **63**(6) 1673–1695.

Burtch, Gordon, Qinglai He, Yili Hong, Dokyun Lee. 2022. How do peer awards motivate creative content? experimental evidence from reddit. *Management Science* **68**(5) 3488–3506.

Burtch, Gordon, Yili Hong, Ravi Bapna, Vladas Griskevicius. 2018. Stimulating online reviews by combining financial incentives and social norms. *Management Science* **64**(5) 2065–2082.

Burtch, Gordon, Dokyun Lee, Zhichen Chen. 2023. Generative ai for ugc and online community engagement. *Available at SSRN 4521754* .

Cascio Rizzo, Giovanni Luca, Jonah A. Berger, Mi Zhou. 2024. Talking with your hands: How hand gestures influence communication. *The Wharton School Research Paper* URL https://ssrn.com/abstract=4962927. Available at SSRN 4962927.

Chen, Jiawei, Luo He, Hongyan Liu, Yinghui Yang, Xuan Bi. 2024. Background music recommendation on short video sharing platforms. *Information Systems Research* .

Chen, Zenan, Jason Chan. 2023. Large language model in creative work: The role of collaboration modality and user expertise. *Management Science* .

Cheng, Zhaoqi, Dokyun Lee, Prasanna Tambe. 2022. Innovae: Generative ai for mapping patents and firm innovation. *Available at SSRN 3868599* .

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, James Robins. 2018. Double/debiased machine learning for treatment and structural parameters.

Choi, Tsan-Ming, Stein W Wallace, Yulan Wang. 2018. Big data analytics in operations management. *Production and Operations Management* **27**(10) 1868–1883.

Clyde, Nicholas, Dennis Zhang, Bing Bai. 2024. The impact of ridesharing platforms on healthcare access. *Available at SSRN 4968892* .

Coates, Jennifer. 2015. *Women, men and language: A sociolinguistic account of gender differences in language*. Routledge.

Cole, Rebel A, Tatyana Sokolyk. 2018. Debt financing, survival, and growth of start-up firms. *Journal of Corporate Finance* **50** 609–625.

Cui, Ruomeng, Jun Li, Dennis J. Zhang. 2020a. Reducing discrimination with reviews in the sharing economy: Evidence from field experiments on airbnb. *Management Science* **66**(3) 1071–1094.

Cui, Ruomeng, Meng Li, Qiang Li. 2020b. Value of high-quality logistics: Evidence from a clash between sf express and alibaba. *Management Science* **66**(9) 3879–3902.

Cui, Ruomeng, Meng Li, Shichen Zhang. 2022. Ai and procurement. *Manufacturing & Service Operations Management* **24**(2) 691–706.

Davidson, James, Benjamin Liebald, Junning Liu, Palash Nandy, Taylor Van Vleet, Ullas Gargi, Sujoy Gupta, Yu He, Mike Lambert, Blake Livingston, et al. 2010. The youtube video recommendation system. *Proceedings of the fourth ACM conference on Recommender systems*. 293–296.

Dukes, Anthony, Qihong Liu. 2024. The consumption of advertising in the digital age: Attention and ad content. *Management Science* **70**(4) 2086–2106.

Ellis, Scott C., Shashank Rao, Dheeraj Raju, Thomas J. Goldsby. 2018. Rfid tag performance: Linking the laboratory to the field through unsupervised learning. *Production and Operations Management* **27**(10) 1834–1848.

Fang, Zhen, Ming Fan, Apurva Jain. 2023. Content proliferation and narrowcasting in the age of streaming media. *Production and Operations Management* **32**(10) 3295–3310.

Goes, Paulo B, Mingfeng Lin, Ching-man Au Yeung. 2014. "popularity effect" in user-generated content: Evidence from online product reviews. *Information Systems Research* **25**(2) 222–238.

Golubovikj, Arsen Matej, Marko Tkalčič. 2025. Rethinking subjective features in recommender systems: Personal views over aggregated values. *Proceedings of the Nineteenth ACM Conference on Recommender Systems*. 1279–1283.

Granulo, Armin, Christoph Fuchs, Stefano Puntoni. 2021. Preference for human (vs. robotic) labor is stronger in symbolic consumption contexts. *Journal of Consumer Psychology* **31**(1) 72–80.

Guo, Yutong, Ying Zhang, Khim-Yong Goh, Xixian Peng. 2024. Can social technologies drive purchases in e-commerce live streaming? an empirical study of broadcasters' cognitive and affective social call-to-actions. *Production and Operations Management* 10591478241276131.

He, Qinglai, Yili Hong, TS Raghu. 2021. The effects of machine-powered platform governance: An empirical study of content moderation. *Available at SSRN 3767680* .

Hoiles, William, Anup Aprem, Vikram Krishnamurthy. 2017. Engagement and popularity dynamics of youtube videos and sensitivity to meta-data. *IEEE Transactions on Knowledge and Data Engineering* **29**(7) 1426–1437.

Huang, Ni, Gordon Burtch, Bin Gu, Yili Hong, Chen Liang, Kanliang Wang, Dongpu Fu, Bo Yang. 2019. Motivating user-generated content with performance feedback: Evidence from randomized field experiments. *Management Science* **65**(1) 327–345.

Huang, Ni, Probal Mojumder, Tianshu Sun, Jinchi Lv, Joseph M Golden. 2021. Not registered? please sign up first: A randomized field experiment on the ex ante registration request. *Information Systems Research* **32**(3) 914–931.

Kesavan, Saravanan, Tarun Kushwaha. 2020. Field experiment on the profit implications of merchants' discretionary power to override data-driven decision-making tools. *Management Science* **66**(11) 5182–5190.

Kuang, Lini, Ni Huang, Yili Hong, Zhijun Yan. 2019. Spillover effects of financial incentives on non-incentivized user engagement: Evidence from an online knowledge exchange platform. *Journal of Management Information Systems* **36**(1) 289–320.

Lee, Heeseung Andrew, Angela Aerry Choi, Wonseok Oh, Tianshu Sun. 2025. To split or to merge? how partitioning affects consumption and engagement with digital content. *Information Systems Research* .

Li, Yiyi, Ying Xie. 2020. Is a picture worth a thousand words? an empirical study of image content and social media engagement. *Journal of marketing research* **57**(1) 1–19.

Liang, Paul Pu, Chiyu Wu, Louis-Philippe Morency, Ruslan Salakhutdinov. 2021. Towards understanding and mitigating social biases in language models. *International Conference on Machine Learning*. PMLR, 6565–6576.

Liu, Jin, Xingchen Xu, Xi Nan, Yongjun Li, Yong Tan. 2024. "generate" the future of work through ai: Empirical evidence from online labor markets. URL https://arxiv.org/abs/2308.05201.

Longoni, Chiara, Andrea Bonezzi, Carey K Morewedge. 2019. Resistance to medical artificial intelligence. *Journal of Consumer Research* **46**(4) 629–650.

Lysyakov, Mikhail, Siva Viswanathan. 2023. Threatened by ai: Analyzing users' responses to the introduction of ai in a crowd-sourcing platform. *Information Systems Research* **34**(3) 1191–1210.

Malik, Haroon, Zifeng Tian. 2017. A framework for collecting youtube meta-data. *Procedia Computer Science* **113** 194–201.

Panniello, Umberto, Michele Gorgoglione, Alexander Tuzhilin. 2016. Research note—in carss we trust: How context-aware recommendations affect customers' trust and other business performance measures of recommender systems. *Information Systems Research* **27**(1) 182–196.

Peng, Jiaxu, Jungpil Hahn, Ke-Wei Huang. 2023. Handling missing values in information systems research: A review of methods and assumptions. *Information Systems Research* **34**(1) 5–26.

Qiao, Dandan, Shun-Yang Lee, Andrew B Whinston, Qiang Wei. 2020. Financial incentives dampen altruism in online prosocial contributions: A study of online reviews. *Information Systems Research* **31**(4) 1361–1375.

Reisenbichler, Martin, Thomas Reutterer, David A Schweidel, Daniel Dan. 2022. Frontiers: Supporting content marketing with natural language generation. *Marketing Science* **41**(3) 441–452.

Saar-Tsechansky, Maytal, Prem Melville, Foster Provost. 2009. Active feature-value acquisition. *Management Science* **55**(4) 664–684.

Su, Yi, Qili Wang, Liangfei Qiu, Runyu Chen. 2024. Unveiling the effects of introducing ai-generated summaries in e-commerce. *Available at SSRN 4872205* .

Sun, Jiankun, Dennis J. Zhang, Haoyuan Hu, Jan A. Van Mieghem. 2022. Predicting human discretion to adjust algorithmic prescription: A large-scale field experiment in warehouse operations. *Management Science* **68**(2) 846–865. doi:10.1287/mnsc.2021.3990.

Sun, Tianshu, Lanfei Shi, Siva Viswanathan, Elena Zheleva. 2019. Motivating effective mobile app adoptions: Evidence from a large-scale randomized field experiment. *Information Systems Research* **30**(2) 523–539.

Susarla, Anjana, Ram Gopal, Jason Bennett Thatcher, Suprateek Sarker. 2023. The janus effect of generative ai: Charting the path for responsible conduct of scholarly activities in information systems. *Information Systems Research* **34**(2) 399–408.

Wang, Wen, Siqi Pei, Tianshu Sun. 2025. Unraveling generative ai from a human intelligence perspective: A battery of experiments. *Information System Research* .

Wang, Zhihan (Helen), Jun Li, Di (Andrew) Wu. 2023. Mind the gap: Gender disparity in online learning platform interactions. *Manufacturing & Service Operations Management* **25**(6) 2122–2141.

Wasko, Molly McLure, Samer Faraj. 2005. Why should i share? examining social capital and knowledge contribution in electronic networks of practice. *MIS Quarterly* 35–57.

Wei, Wei, Xubin Ren, Jiabin Tang, Qinyong Wang, Lixin Su, Suqi Cheng, Junfeng Wang, Dawei Yin, Chao Huang. 2024. Llmrec: Large language models with graph augmentation for recommendation. *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*. 806–815.

Ye, Zikun, Dennis J Zhang, Heng Zhang, Renyu Zhang, Xin Chen, Zhiwei Xu. 2023. Cold start to improve market thickness on online advertising platforms: Data-driven algorithms and field experiments. *Management Science* **69**(7) 3838–3860.

Zeng, Zhiyu, Hengchen Dai, Dennis J. Zhang, Heng Zhang, Renyu Zhang, Zhiwei Xu, Zuo-Jun Max Shen. 2023. The impact of social nudges on user-generated content for social network platforms. *Management Science* **69**(9) 5189–5208.

Zhang, Xingyue, James A Dearden, Yuliang Yao. 2022. Let them stay or let them go? online retailer pricing strategy for managing stockouts. *Production and Operations Management* **31**(11) 4173–4190.

Zhou, Eric, Dokyun Lee. 2024. Generative artificial intelligence, human creativity, and art. *PNAS Nexus* **3**(3) 052.

Online Appendices

A. Overview of the Generative AI Markets

Table A.1 Generative AI Market Overview by Modality

Input \ Output	Text	Image	Video
Text	Video script generation (ChatGPT, Kapwing, TikTok Creative Assistant)	Ad/marketing visuals (Midjourney, Adobe Firefly, Ideogram, DALL·E 3)	Ad video generation (Runway Gen-4, Pika, Luma Dream Machine, OpenAI Sora)
Image	Metadata generation (Google Cloud Vision, Azure AI Vision, AWS Textract)	Image re-stylization (Adobe Firefly, Canva Magic Edit, Picsart AI)	Image-to-video animation (Runway Gen-4, Pika, Kaiber.ai, LeiaPix)
Video	Metadata generation (vidIQ, TubeBuddy, Opus Clip, ThinkMetadataAI)	Thumbnail generation (Adobe Express, Canva, vidIQ, Fotor)	Video re-stylization (Runway Gen-1, DeOldify, Topaz Video AI, Krea.ai)

Table A.2 Examples of GAI Tools for Video Metadata Generation

Company	Tool Name	Data Input
ThinkAnalytics	ThinkMetadataAI	Video
Quickplay	AI Studio	Video
Coactive	AI Metadata Enrichment	Video
PodClips	AI Video Metadata Generator	Video (TikTok)
Predis.ai	Predis AI Platform	Video (TikTok)
n8n.io	AI Metadata Workflow	Video (YouTube)
vidIQ	AI Video Title Generator	Video (YouTube)
TubeBuddy	YouTube Title Generator	Video (YouTube)

B. Recommender System Workflow on Platform A

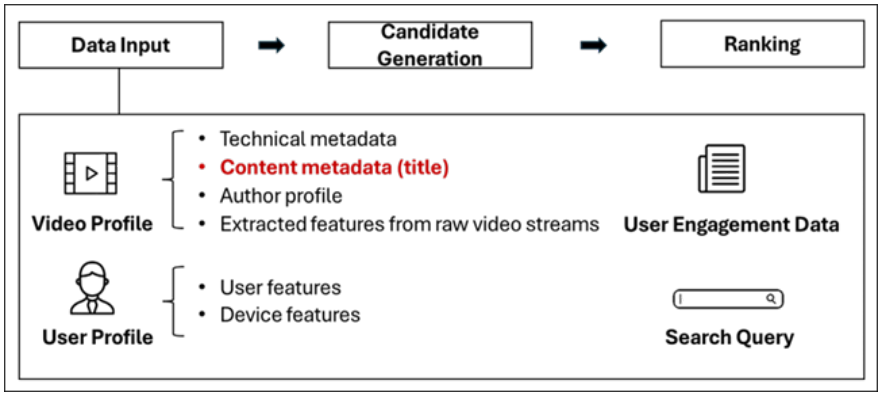


Figure B.1 Recommender System Workflow on Platform A

C. Human- and AI-Generated Title Processes

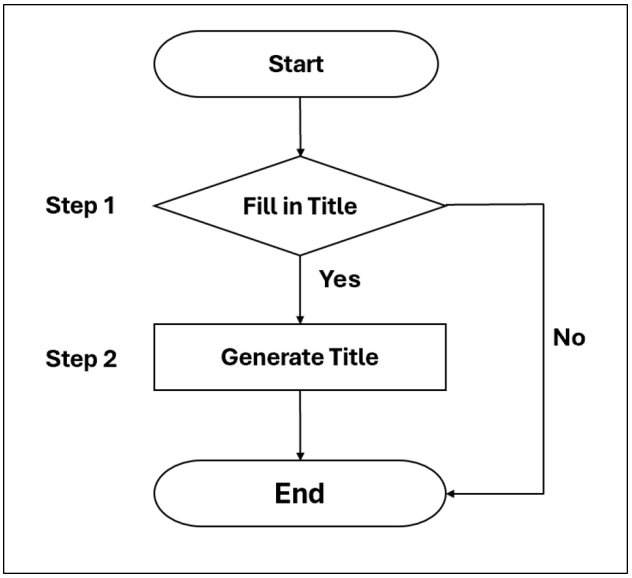


Figure C.1 Human-Generated Title Process

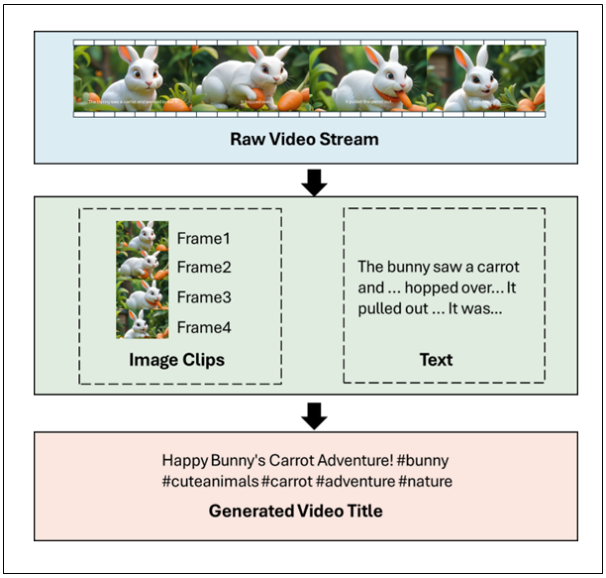


Figure C.2 AI-Generated Title Process

D. Variable Definition and Summary Statistics of Focal Variables

Table D.1 Variable Definitions

Variables	Description
$Treat_i$	Coded as 1 if producer i was assigned to the treatment group, or else as 0.
$ValidWatch_{ij}$	Number of valid watches for video j of producer i.
$WatchDuration_{ij}$	Viewers' total watch duration (in minutes) for video j of producer i.
$Utilitarian_{ij}$	Coded as 1 if video j of producer i is a utilitarian-content video, or else as 0.
$Follower_i$	Number of users that follow producer i.
KOL_i	Coded as 1 if producer i is a key opinion leader, or else as 0.
$Experience_i$	Tenure of producer i (in years) on the platform.
$Following_i$	Number of users that producer i follows.
$Multihome_i$	Coded as 1 if producer i multihomes on other short video platforms, or else as 0.
$Female_i$	Coded as 1 if producer i is female, or else as 0.
$Province_i$	Province location dummies for producer i.
$PublicVisible_{ij}$	Coded as 1 if video j is publicly visible.
$VideoDuration_{ij}$	Duration of video j (in minutes).
$Cover_{ij}$	Coded as 1 if producer i manually sets a video cover for video j.
$ContentType_{ij}$	Coded as 1 if video j is composed of videos (i.e., not images), or else as 0.
$PostDate_{ij}$	Video posting date dummies.
$Category_{ij}$	First-level category dummies for video j of producer i.

Notes: All variables are coded as described in the table. Video metadata variables are collected from the platform's system logs. The variables $WatchDuration_{ij}$ and $ValidWatch_{ij}$ measure engagement metrics.

Table D.2 Summary Statistics of Focal Variables

Variables	Mean	SD	Min	Max
Treat	0.501	0.500	0	1
ValidWatch	236.543	8,718.901	0	10,504,166
WatchDuration	249.240	10,995.043	0	16,021,141
Utilitarian	0.067	0.251	0	1
Follower	477.066	649.083	0	5,133
KOL	0.005	0.073	0	1
Experience	2.209	1.763	0.003	5.631
Following	2,940.926	50,989.248	0	37,260,268
Multihome	0.732	0.443	0	1
Female	0.649	0.477	0	1
PublicVisible	0.921	0.270	0	1
VideoDuration	0.466	1.020	0	43.532
Cover	0.137	0.344	0	1
ContentType	0.744	0.437	0	1

Notes: All variables are calculated based on video-level and producer-level data. SD stands for standard deviation, and Min and Max represent the minimum and maximum values observed for each variable. Values for *ValidWatch*, *WatchDuration*, *Follower*, and *Following* have been scaled.

E. Pearson Correlation Matrix of Focal Variables

Table E.1 Pearson Correlation Matrix of Focal Variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) *Treat*	1											
(2) *ValidWatch*	0.00	1										
(3) *WatchDuration*	0.00	0.80	1									
(4) *Utilitarian*	0.01	0.01	0.01	1								
(5) *Follower*	0.00	-0.17	0.13	0.01	1							
(6) *KOL*	0.00	0.02	0.01	0.01	0.12	1						
(7) *Experience*	0.00	0.00	0.00	-0.05	0.04	0.06	1					
(8) *Following*	0.00	0.01	-0.01	0.02	-0.01	0.00	0.15	1				
(9) *Multihome*	0.00	0.01	0.01	-0.01	0.02	0.03	0.11	-0.05	1			
(10) *Female*	0.00	-0.01	-0.01	-0.03	-0.01	-0.03	0.04	-0.03	0.06	1		
(11) *PublicVisible*	0.00	0.01	0.01	0.03	0.01	0.02	-0.03	0.06	-0.01	-0.09	1	
(12) *VideoDuration*	0.00	0.03	0.04	0.05	0.03	0.01	0.06	0.05	0.00	-0.06	0.02	1

F. Randomization Check Results

Table F.1 Randomization Check Results

Variables	Treatment Producers	Control Producers	P-value of t-test
ValidWatch	163.055	176.388	0.248
WatchDuration	137.386	146.698	0.453
Utilitarian	0.065	0.066	0.238
Follower	1286.531	1261.090	0.601
KOL	0.003	0.003	0.830
Experience	2.343	2.341	0.553
Following	378.537	378.720	0.817
Multihome	0.856	0.855	0.103
Female	0.610	0.610	0.597
PublicVisible	0.940	0.941	0.259
VideoDuration	0.465	0.465	0.897
Cover	0.155	0.156	0.135
ContentType	0.757	0.757	0.725

Notes: The p-value column represents the significance level from a t-test comparing the treatment and control groups. Values for *ValidWatch*, *WatchDuration*, *Follower*, and *Following* have been scaled.

G. Results for Video Production Comparison

We examined whether access to AI-generated titles affected producers' video production (e.g., producing more or fewer videos). To test this, we conducted producer-level t-tests comparing (1) the total number of videos produced and (2) the average time gaps (in hours) between consecutive videos for treatment and control groups during the experiment. As shown in Table G.1, neither measure shows a statistically significant difference. These results suggest that the provision of AI-generated titles primarily affects video titling rather than video production. This finding is consistent with the design of the intervention, as title generation occurs after video creation and immediately before posting, leaving little scope for a direct influence on production.

Table G.1 Video Production Comparison

Variables	Treatment Group	Control Group	Difference	P-value
NumVideo	5.246	5.240	0.006	0.660
TimeGap	51.241	51.249	-0.008	0.928

H. Model Estimation Results for LATE

Considering that about 60% of videos in the treatment group did not have AI-generated titles populated in the title-setting box, we further estimated the local average treatment effect (LATE) to capture the impact on those videos that were effectively influenced by the provision of AI-generated titles (i.e., Panels C and D in Figure 3). To identify this effect, we cannot simply compare videos with and without AI-generated titles populated. This is because unobserved factors, such

as producers' own titling capabilities or effort levels, may simultaneously affect the likelihood of populating AI-generated titles and subsequent video viewership outcomes. To address this concern, we used the random assignment of producers to the treatment group ($Treat_i$) as an instrumental variable (IV) for the populated decision (Huang et al. 2021, Sun et al. 2019). We employed the following two-stage least squares (2SLS) regression:

$$Populate_{ij} = \gamma_0 + \gamma_1 Treat_i + \gamma_2 Controls_{ij} + \epsilon_{ij} \tag{H.1}$$

$$Outcome_{ij} = \lambda_0 + \lambda_1 \widehat{Populate}_{ij} + \lambda_2 Controls_{ij} + \eta_{ij} \tag{H.2}$$

where ϵ_{ij} and η_{ij} are error terms. In Equation (H.1), $Populate_{ij}$ is a binary indicator for whether the AI-generated title was populated in the title-setting box for video j of producer i, and is instrumented with $Treat_i$. In Equation (H.2), $\widehat{Populate}_{ij}$ refers to the instrumented value of $Populate_{ij}$, i.e., the fitted value obtained from Equation (H.1). The coefficient of interest, λ_1, captures the local average treatment effect (LATE) of having the AI-generated title populated. $Treat_i$ is a valid IV for two reasons. First, it satisfies the relevance assumption as only producers in the treatment group had access to AI-generated titles, which strongly predicts the platform's AI-generated title population decisions. This is evidenced by the very large first-stage F-statistic (3.7×10^6). Second, it satisfies the exclusion restriction because the treatment assignment is random and should not correlate with other observed or unobserved covariates. Moreover, title generation occurs after video upload and just before posting, removing direct influence on video production.

Table H.1 presents the estimation results. Treatment group videos that had AI-generated titles populated in the title-setting box increased valid watches by 4.0% (p-value < 0.01) and watch duration by 2.3% (p-value < 0.05). These results are consistent with our findings in the main text (see Section 5.1) but show a larger effect size, reflecting the attenuation in the earlier estimates caused by the substantial fraction of treatment videos where AI-generated titles were not populated.

Table H.1 Model Estimation Results for LATE

Variables	*ValidWatch*	*WatchDuration*
	(1)	(2)
Populate	**0.040*****	**0.023****
	(0.011)	(0.010)
Controls	Yes	Yes
Observations	10,738,984	10,738,984
R-squared	0.296	0.322

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors clustered at the producer level.

I. Title Visibility

We examined the title-visibility mechanism by testing whether the treatment effect diminished as videos accumulated more valid watches. Each additional valid watch provides the platform with another observed viewer-video interaction data, allowing the recommender system to learn which users are likely to engage with the focal video. If title visibility were the primary mechanism, the treatment effect should remain relatively stable as a video accumulates more interactions, because title visibility does not depend on how much the system has already learned about that video. By contrast, if improved user-video matching were the main mechanism, the marginal benefit of metadata should decline over time, because the recommender system can increasingly rely on accumulated interaction data rather than metadata to identify suitable viewers. To test these competing explanations, we sampled the videos published on August 8th, 2023, and tracked their performance for 14 days after the video was posted. We estimated the following model specification:

$$Outcome2_{ijt} = \gamma_0 + \gamma_1\, Treat_i + \gamma_2 \left(Treat_i \times AccuVW_{ijt}\right) + \gamma_3\, Controls_{ijt} + \tau_t + \epsilon_{ijt} \tag{I.1}$$

where $Outcome2_{ijt}$ denotes our two video consumption outcomes for video j of user i on day t, and $AccuVW_{ijt}$ is the accumulated number of valid watches for video j up to day t. τ_t represents time fixed effects that absorb temporal shocks, and ϵ_{ijt} is the error term. Both $Outcome2_{ijt}$ and $AccuVW_{ijt}$ as well as all highly-skewed control variables, are log-transformed. The focal model coefficient of interest is γ_2. Our estimation results in Table I.1 show a significantly negative coefficient for the interaction term between $Treat_i$ and $AccuVW_{ijt}$, suggesting that accumulated viewership negatively moderates the relationship between AI-generated titles and video consumption outcomes. Therefore, the positive effects of AI-generated titles on video viewership should be attributed to better user-video matching, instead of title visibility.

Table I.1 Diminishing Treatment Effects with Accumulated Video Consumption

Variables	*ValidWatch* (1)	*WatchDuration* (2)
Treat	0.006**	0.007***
	(0.002)	(0.002)
AccuVW	0.477***	0.395***
	(0.0005)	(0.0005)
Treat × *AccuVW*	**-0.003*****	**-0.004*****
	(0.001)	(0.001)
Controls	Yes	Yes
Observations	7,288,648	7,288,648
R-squared	0.615	0.594

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors.

J. Algorithmic Promotion

Another potential explanation for the positive treatment effect is algorithmic promotion. The platform might have intentionally boosted videos with AI-generated titles (e.g., to encourage adoption of the new tool), rather than the effect arising from improved metadata. To test whether algorithmic promotion is a driver for the effect of AI-generated titles, we normalized key outcomes (e.g., valid watch count and watch duration) by total video views and re-estimated Equation (2). The treatment effects, shown in Table J.1, remain statistically significant, reinforcing that the improvements should not be attributed to algorithmic promotion. We consulted Platform A's recommendation algorithm team, which confirmed that adopting AI-generated titles did not trigger any additional prioritization in the recommender system. This suggests that the observed increase in video viewership is unlikely a result of algorithmic promotion.

Table J.1 Model Estimation Results of Treatment Effects on Normalized Viewership Outcomes

Variables	*ValidWatch* (1)	*WatchDuration* (2)
Treat	**0.003*****	**0.001*****
	(0.0004)	(0.0003)
Controls	Yes	Yes
Observations	10,695,530	10,695,530
R-squared	0.020	0.031

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors clustered at the producer level. 43,454 observations were excluded due to zero total video views.

K. Viewer-Side Cannibalization on Control Group Videos

An alternative explanation is that treatment group videos could crowd out viewers' engagement with control group videos. We argue that such large-scale cannibalization is unlikely, as treatment producers represent only about 1% of all producers on Platform A. To empirically verify this, we collected a new dataset that tracks the daily viewing records of a random sample of 10,000 viewers who watched at least one treatment or control video. Using this user-day-level dataset, we estimated the following model with robust standard errors:

$$ControlVW_{it} = \beta_0 + \beta_1\, TreatVideoShare_{it} + \beta_2\, Controls_{it} + \mu_i + \delta_t + \varepsilon_{it}. \tag{K.1}$$

where $ControlVW_{it}$ is the number of valid watches of control group videos by user i on day t, and $TreatVideoShare_{it}$ is the share of treatment group videos watched by the user that day. μ_i denotes the viewer fixed effects and δ_t represents day fixed effects. The focal coefficient of

interest is β_1. Results in Table K.1 show that the coefficient of *TreatVideoShare* is positive and statistically significant, indicating that exposure to treatment content did not suppress control group consumption. Instead, users who watched more treatment group videos also engaged more with control group videos. A plausible explanation is that users exposed to treatment videos experienced better user-video matching and greater satisfaction, which in turn motivated them to consume additional videos. Taken together, this evidence rules out the concern that treatment group videos outperformed control group videos by crowding out control group videos.

Table K.1 Effect of Treatment Video Exposure on Control Group Consumption

Variables	*ControlVW*
	(1)
TreatVideoShare	**1.397*****
	(0.017)
Controls	Yes
Observations	450,609

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors clustered at the producer level.

L. Propensity Score Matching (PSM) Results for Titled Videos

From the total of 10,738,984 videos, we first excluded 3,444,235 videos that did not have titles from both the treatment and control groups (Panels A+C+E in Figure 3). Additionally, we removed 2,226,922 titled videos (Panel F in Figure 3) in our treatment group that did not receive AI-generated titles due to algorithmic limitations. The filtered sample consisted of 2,092,219 videos in the treatment group (Panel D in Figure 3) and 2,975,608 in the control group (Panel B in Figure 3). Next, we used the Propensity Score Matching (PSM) method to identify a sample that was similar in observed characteristics during the pre-treatment period. The matching followed a two-step procedure: first, we ran a logit regression using the pre-treatment variables (i.e., all the moderating and control variables mentioned in Section 3.3) and obtained predicted propensity scores for each unit. Second, we employed a one-to-many radius matching algorithm where all control units for which the propensity scores fall within a pre-defined radius (also known as caliper) from the propensity scores of the treatment units are matched. This ensures multiple matches for each treated unit. We then applied weights in the subsequent analysis to account for this one-to-many structure. Next, we obtained a new sample after discarding unmatched units (3,885,089 videos were matched). To evaluate the matching quality, we performed t-tests of equality of means before and after matching to verify whether our matching had successfully balanced the attributes between the treatment and control group videos. The results in Table L.1 show that the mean differences between the groups were no longer statistically significant, indicating that the

matching procedure successfully reduced differences of observed covariates across treatment and control groups.

Table L.1 Differences in Mean Before and After Matching (PSM, Radius Matching)

Variables	Sample	Mean Treated	Mean Control	%Bias	T-statistics	P-value
Utilitarian	Before-matched	0.065	0.069	-1.900	-20.650	0.000
	After-matched	0.065	0.073	-3.400	-0.050	0.960
Ln(*Follower*)	Before-matched	2.975	2.710	28.700	315.750	0.000
	After-matched	2.975	2.905	3.800	0.110	0.911
KOL	Before-matched	0.004	0.008	-4.500	-48.470	0.000
	After-matched	0.004	0.005	-1.100	-0.020	0.986
Experience	Before-matched	2.642	2.072	31.900	354.270	0.000
	After-matched	2.642	2.505	7.700	0.110	0.915
Ln(*Following*)	Before-matched	2.907	2.989	-8.200	-89.620	0.000
	After-matched	2.907	2.881	2.600	0.040	0.968
Multihome	Before-matched	0.720	0.768	-10.900	-121.070	0.000
	After-matched	0.720	0.735	-3.400	-0.050	0.962
Female	Before-matched	0.658	0.658	-0.100	-1.500	0.000
	After-matched	0.658	0.649	1.900	0.030	0.979
PublicVisible	Before-matched	0.921	0.949	-11.200	-126.320	0.000
	After-matched	0.921	0.930	-3.400	-0.040	0.965
Ln(*VideoDuration*)	Before-matched	0.327	0.269	16.000	177.980	0.000
	After-matched	0.327	0.337	-2.600	-0.040	0.971
Cover	Before-matched	0.135	0.196	-16.300	-178.890	0.000
	After-matched	0.135	0.160	-6.700	-0.100	0.918
ContentType	Before-matched	0.828	0.660	39.200	425.900	0.000
	After-matched	0.828	0.850	-5.100	-0.080	0.934

Notes: ***p <0.01; **p <0.05; *p <0.1. %Bias measures the standardized difference in covariate means, scaled by the standard deviation of the sample including both treatment and control groups. The closer it is to zero, the better the balance between the treatment and control groups. Values for *Follower* and *Following* have been scaled.

M. Validation of the Similarity Measure

To validate that the similarity score meaningfully captures producers' revision behaviors, we conducted a complementary analysis. Although the AI-generated title algorithm used during the experimental period is no longer available, the platform has since deployed an updated AI-generated hashtag tool based on a comparable text-generation framework. We used this GAI tool to generate hashtags for control group videos and compared them with human-generated hashtags extracted from the corresponding titles. The average similarity score was 0.031, and even by category, the values ranged from 0.01 to 0.077, which are consistently small. These results indicate that AI-generated titles differ substantially from human-generated ones, suggesting that high similarity scores reflect producers' direct adoption of AI-generated titles, while low scores indicate meaningful revision. We acknowledge that some producers with very low similarity scores may have fully deleted the AI-generated titles and replaced them with new ones, rather than substantially revising them. Accordingly, low similarity should be interpreted more broadly as indicating greater departure from the original AI-generated title, rather than revision alone.

N. Alternative Identification Strategies for Videos with Low Textual Similarity

A potential endogeneity concern is that the videos with significantly revised AI-generated titles, resulting in lower similarity between the AI-generated titles and actual titles adopted by producers, may inherently indicate higher producer effort and video quality. In other words, the observed outcomes could stem from these underlying differences rather than the benefits of human-AI co-creation. To address this issue, from the 2,092,219 titled videos (Panel D in Figure 3) in the treatment group and 2,975,608 in the control group (Panel B in Figure 3), we first excluded 1,532,819 treatment videos with cosine similarity to AI-generated titles higher than 20%. This filtering resulted in 555,066 videos in the treatment group and 2,975,608 in the control group.

Based on this sample, we employed two identification strategies. First, we applied inverse probability weighting (IPW) on the full set of treatment and control videos, re-weighting observations according to estimated propensity scores. Second, we employed the propensity score matching (PSM) method using a radius matching algorithm to construct a one-to-many matched sample based on pre-treatment covariates (i.e., all moderating and control variables in Section 3.3), following the procedure detailed in Online Appendix L. After discarding unmatched units, the final matched sample included 3,533,720 videos. Post-matching t-tests of equality of means, shown in Table N.1, indicate that the mean differences between the treatment and control groups were no longer statistically significant, confirming that the matching procedure was effective. For both approaches, we re-estimated Equation (2) using the weighted or matched samples. The results in Table N.2 show that videos in the treatment group outperformed those in the control group in terms of valid watches and watch duration. These findings suggest that the observed higher viewership outcomes are likely driven by human-AI co-creation.

Table N.1 Differences in Mean Before and After Matching (PSM, Radius Matching)

Variables	Sample	Mean Treated	Mean Control	%Bias	T-statistics	P-value
Utilitarian	Before-matched	0.065	0.069	-1.900	-20.650	0.000
	After-matched	0.066	0.069	-1.100	-0.040	0.966
Ln(*Follower*)	Before-matched	2.975	2.710	28.700	315.750	0.000
	After-matched	2.716	2.700	1.700	0.070	0.948
KOL	Before-matched	0.004	0.008	-4.500	-48.470	0.000
	After-matched	0.006	0.007	-1.400	-0.060	0.955
Experience	Before-matched	2.642	2.072	31.900	354.270	0.000
	After-matched	2.541	2.075	25.600	0.930	0.353
Ln(*Following*)	Before-matched	2.907	2.989	-8.200	-89.620	0.000
	After-matched	2.911	2.947	-3.600	-0.140	0.886
Multihome	Before-matched	0.720	0.768	-10.900	-121.070	0.000
	After-matched	0.777	0.767	2.600	0.100	0.922
Female	Before-matched	0.658	0.658	-0.100	-1.500	0.000
	After-matched	0.660	0.657	0.700	0.030	0.980
PublicVisible	Before-matched	0.921	0.949	-11.200	-126.320	0.000
	After-matched	0.959	0.952	3.200	0.130	0.898
Ln(*VideoDuration*)	Before-matched	0.327	0.269	16.000	177.980	0.000
	After-matched	0.277	0.263	4.000	0.150	0.881
Cover	Before-matched	0.135	0.196	-16.300	-178.890	0.000
	After-matched	0.226	0.194	7.800	0.280	0.777
ContentType	Before-matched	0.828	0.660	39.200	425.900	0.000
	After-matched	0.648	0.647	0.200	0.010	0.994

Notes: $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.1$. %Bias measures the standardized difference in covariate means, scaled by the standard deviation of the sample including both treatment and control groups. The closer it is to zero, the better the balance between the treatment and control groups. Values for *Follower* and *Following* have been scaled.

Table N.2 Model Estimation Results for Titled Videos with Lower Textual Similarity

	IPW-Weighted Sample		PSM-Matched Sample	
Variables	*ValidWatch*	*WatchDuration*	*ValidWatch*	*WatchDuration*
	(1)	(2)	(3)	(4)
Treat	**0.151*****	**0.115*****	**0.153*****	**0.121*****
	(0.003)	(0.003)	(0.002)	(0.002)
Relative Effect Size	16.3%	12.2%	16.5%	12.9%
Controls	Yes	Yes	Yes	Yes
Observations	3,885,089	3,885,089	3,533,720	3,533,720
R-squared	0.261	0.286	0.242	0.269

Notes: $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$. Values in parentheses are robust standard errors. The number of observations differs across methods because IPW retains the full sample while PSM restricts to the matched subsample.

O. Alternative Identification: Double Machine Learning (DML) Approach

In Section 6.2, we employed propensity score matching (PSM) to mitigate bias from observable differences between treatment and control videos. To test the robustness of our results, we adopted an alternative identification strategy based on double machine learning (DML) (Chernozhukov et al. 2018). The DML framework leverages machine learning methods to flexibly capture

high-dimensional and nonlinear confounding while estimating the linear relationship between our focal variables and outcomes. Due to computational constraints, we could not implement DML on the entire PSM-matched dataset[1]. Instead, we randomly selected a 1% subsample (38,499 videos) from the matched sample. Table O.1 reports a summary statistics of the full matched sample, and Table O.2 reports those of the 1% subsample. The distributions are highly comparable, indicating that the subsample is representative.

Table O.1 Summary Statistics of Focal Variables for PSM-Matched Sample

Variables	Mean	SD	Min	Max
Treat	0.539	0.499	0	1
ValidWatch	237.7	9,339	0	3,410,247
WatchDuration	187.4	10,134	0	5,084,853
Utilitarian	0.069	0.253	0	1
Follower	990.9	1,318	0	5,133
KOL	0.005	0.071	0	1
Experience	2.493	1.829	0.003	5.631
Following	3,237	58,359	0	24,394,694
Multihome	0.730	0.444	0	1
Female	0.653	0.476	0	1
PublicVisible	0.927	0.259	0	1
VideoDuration	0.513	1.015	0	35.400
Cover	0.152	0.359	0	1
ContentType	0.821	0.383	0	1

Notes: SD = standard deviation. All variables at video or producer level.

Table O.2 Summary Statistics for 1% Subsample of PSM-Matched Sample

Variable	Mean	SD	Min	Max
Treat	0.536	0.499	0	1
ValidWatch	210.2	7,149	0	913,791
WatchDuration	204.3	11,507	0	1,677,096
Utilitarian	0.070	0.254	0	1
Follower	983.8	1,314	0	5,069
KOL	0.006	0.077	0	1
Experience	2.486	1.824	0.003	5.631
Following	3,174	51,895	0	7,578,791
Multihome	0.729	0.445	0	1
Female	0.652	0.476	0	1
PublicVisible	0.926	0.261	0	1
VideoDuration	0.513	1.014	0	23.270
Cover	0.151	0.358	0	1
ContentType	0.817	0.387	0	1

Notes: SD = standard deviation. All variables at video or producer level.

We employed the DML framework (via Stata's *ddml* package) to calibrate the partially linear model (i.e., PLM model) (Ahrens et al. 2024), as specified in Equations (O.1a)-(O.1c), to estimate the heterogeneous effect of textual similarity on viewership.

$$Outcome2_{ij} = \alpha_0 + \alpha_1 Treat_i + \alpha_2 (Treat_i \times Similarity_{ij}) + g_0(X_{ij}) + \mu_{ij} \tag{O.1a}$$

$$Similarity_{ij} = h_0(X_{ij}) + \xi_{ij} \tag{O.1b}$$

$$Treat_i = j_0(X_{ij}) + \tau_{ij} \tag{O.1c}$$

where $Outcome2_{ij}$ refers to either $ValidWatch_{ij}$ and $WatchDuration_{ij}$. X_{ij} denotes the control variables in our main model. $g_0(\cdot)$, $h_0(\cdot)$, and $j_0(\cdot)$ are nonlinear functions estimated using machine learning methods (i.e., gradient boosting and random forest). μ_{ij}, ξ_{ij}, and τ_{ij} are error terms. The focal model coefficients of interest are α_1 and α_2.

[1] We apply DML to the PSM-matched sample rather than the full dataset because identification requires both unconfoundedness and overlap. Matching helps ensure overlap by excluding observations with zero or near-zero weights, making treatment and control groups more comparable.

The model implementation process is as follows. First, we chose either gradient boosting or random forest to estimate $g_0(\cdot)$, $h_0(\cdot)$, and $j_0(\cdot)$. Second, we computed the residuals from Equations (O.1a), (O.1b), and (O.1c), and applied a two-stage least squares (2SLS) regression on the residualized variables to estimate α_1 and α_2. The DML uses cross-fitting to control for the impact of the first-step estimation errors on the second-step estimations. To implement cross-fitting, we randomly split the sample into K evenly-sized folds and set $K = 5$ (Ahrens et al. 2024). The results from Table O.3 show that the focal coefficients are qualitatively consistent with those in our main results presented in Table 11. We then re-estimated Equation (O.1) by replacing $Outcome2_{ij}$ with variables representing lexical richness (*LexicalDensity*, *LexicalVariation*, and *Entropy*). The results, reported in Table O.4, are similar and further validate the robustness of our findings.

Table O.3 DML-Corrected Heterogeneous Estimation Results for Titled Videos (Matched Sample)

Variables	*ValidWatch*		*WatchDuration*	
	Gradient Boosting (1)	Random Forest (2)	Gradient Boosting (3)	Random Forest (4)
Treat	**0.234*****	**0.179*****	**0.199*****	**0.139*****
	(0.021)	(0.021)	(0.020)	(0.020)
Treat × *Similarity*	**-0.826*****	**-0.708*****	**-0.686*****	**-0.578*****
	(0.023)	(0.023)	(0.022)	(0.022)
Controls	Yes	Yes	Yes	Yes
Observations	38,499	38,499	38,499	38,499

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors. Gradient Boosting and Random Forest are implemented as DML learners.

Table O.4 DML-Corrected Results of Lexical Richness Analysis (Matched Sample)

Variables	*LexicalDensity*		*LexicalVariation*		*Entropy*	
	Gradient Boosting (1)	Random Forest (2)	Gradient Boosting (3)	Random Forest (4)	Gradient Boosting (5)	Random Forest (6)
Treat	**0.017*****	**0.013*****	**0.016*****	**0.013*****	**0.032*****	**0.023*****
	(0.002)	(0.002)	(0.002)	(0.002)	(0.005)	(0.005)
Treat × *Similarity*	**-0.277*****	**-0.262*****	**-0.208*****	**-0.195*****	**-0.520*****	**-0.484*****
	(0.002)	(0.002)	(0.003)	(0.003)	(0.006)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	38,499	38,499	38,499	38,499	38,499	38,499

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors. Gradient Boosting and Random Forest are implemented as DML learners.

We similarly employed the DML approach to estimate the heterogeneous effects of human-AI co-creation behaviors on viewership, specified as:

$$Outcome2_{ij} = \ell_0 + \ell_1 ContextRevise_{ij} + \ell_2 EmotionRevise_{ij} + \ell_3 Treat_i + g_1(X_{ij}) + u_{ij} \tag{O.2a}$$

$$ContextRevise_{ij} = h_1(X_{ij}) + v_{ij} \tag{O.2b}$$

$$EmotionRevise_{ij} = h_2(X_{ij}) + w_{ij} \tag{O.2c}$$

$$Treat_i = j_1(X_{ij}) + \tau_{ij} \tag{O.2d}$$

where $Outcome2_{ij}$ represents our two viewership metrics and X_{ij} again denotes the set of control variables. The functions $g_1(\cdot)$, $h_1(\cdot)$, and $h_2(\cdot)$ are estimated using machine learning methods, and u_{ij}, v_{ij}, w_{ij}, and τ_{ij} are error terms. The coefficients of interest are ℓ_1, ℓ_2, and ℓ_3. The results from Table O.5 show that the focal coefficients are qualitatively consistent with those in our main results presented in Table 13.

Table O.5 DML-Corrected Results of Human-AI Co-Creation on Viewership Outcomes

Variables	*ValidWatch*		*WatchDuration*	
	Gradient Boosting (1)	Random Forest (2)	Gradient Boosting (3)	Random Forest (4)
ContextRevise	**0.092*****	**0.086*****	**0.068*****	**0.061*****
	(0.024)	(0.024)	(0.023)	(0.022)
EmotionRevise	**0.113*****	**0.107*****	**0.113*****	**0.100*****
	(0.025)	(0.026)	(0.024)	(0.024)
Adopt	**-0.606*****	**-0.535*****	**-0.497*****	**-0.442*****
	(0.019)	(0.018)	(0.016)	(0.017)
Controls	Yes	Yes	Yes	Yes
Observations	38,499	38,499	38,499	38,499

Notes: Robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

P. Additional Analyses and Robustness Tests

P.1. Producer-Side Contamination via Private Messaging

A potential concern for causal identification is treatment spillover through private messaging between producers, which would raise concerns about the SUTVA. On Platform A, private messaging is the main direct channel for producer-to-producer interaction (Zeng et al. 2023). During the experiment, treated producers may have informed control producers about the availability of AI-generated titles. If so, control producers may have become aware that they were excluded from the treatment condition, which could have altered their subsequent behavior and contaminated the control group. For example, such awareness may have discouraged control producers or affected how they created video content and metadata. To assess this possibility, we conducted two complementary analyses using a supplementary dataset that records private message exchanges.

P.1.1 Excluding Producers with Cross-Group Private Messaging

We first examined whether cross-group private messaging biased our results. To do so, we removed all producers who exchanged private messages with users from the opposite experimental group. This exclusion corresponds to 2,463,639 observations involving 291,042 unique users (143,169 in the treatment group and 147,873 in the control group). We then re-estimated Equation (2) using videos ($N = 8{,}275{,}345$) from the remaining producers. As shown in Table P.1, the coefficient of $Treat$ is qualitatively consistent with our main results, and larger in magnitude. One plausible explanation is treatment effect heterogeneity. The provision of AI-generated titles is more valuable for socially inactive producers, who are less likely to invest effort in creating video titles on their own and may also have more limited knowledge of how to craft effective titles. Overall, these results help rule out the alternative explanation that cross-group private messaging made control group producers aware of the treatment and then led them to change their video content or title metadata.

Table P.1 Model Estimation Results Excluding Producers with Cross-Group Private Messaging

Variables	*ValidWatch*	*WatchDuration*
	(1)	(2)
Treat	**0.035*****	**0.027*****
	(0.005)	(0.004)
Relative Effect Size	3.6%	2.7%
Controls	Yes	Yes
Observations	8,275,345	8,275,345
R-squared	0.305	0.331

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors clustered at the producer level.

P.1.2 Staggered Difference-in-Differences on Message-Linked Control Group Producers

As a second test, we focused on control group producers who privately messaged with treatment group producers during the experiment. Their first cross-group private message may have made them aware of the differential access to AI-generated titles, which affects the subsequent video content or metadata production and ultimately influences video viewership. We therefore examined whether the viewership performance of videos produced after this first private message declined. Following Zeng et al. (2023), we defined the date of a producer's first private message with a treatment user after the experiment began as the onset of potential awareness. We then estimated a staggered difference-in-differences model with doubly robust inverse probability weighting (DR-IPW), implemented via Stata's *csdid* package:

$$Outcome2_{ijt} = \beta_0 + \beta_1\, PrivateMessage_{ijt} + \beta_2\, Controls_{ijt} + \mu_i + \delta_t + \varepsilon_{ijt}. \tag{P.1}$$

where $PrivateMessage_{ijt}$ equals 1 if video j was posted on or after the experiment start date and after producer i's first private message with a treatment group user. The model includes producer fixed effects (μ_i) and day fixed effects (δ_t). β_1 is the focal coefficient of interest. The coefficient of *PrivateMessage* in Table P.2 is statistically insignificant, indicating no systematic change in the performance of videos posted by control group producers after they communicated with treatment group producers. Taken together with the previous test, these results suggest that the main treatment effect is unlikely to be driven by changes in control group producers' video content or metadata production caused by awareness of the experimental manipulation.

Table P.2 Effect of Private Messaging on Content Consumption (Control Group Sample)

Variables	*ValidWatch*	*WatchDuration*
	(1)	(2)
PrivateMessage	**0.025**	**0.025**
	(0.035)	(0.033)
Controls	Yes	Yes
Observations	389,428	389,428

Notes: $***p < 0.01$, $**p < 0.05$, $*p < 0.1$. Values in parentheses are robust standard errors clustered at the producer level. Estimates are obtained using staggered DID with doubly robust inverse probability weighting (DR-IPW), implemented via Stata's `csdid` package.

P.2. Results for Long-Term Effect Analyses

One may be concerned that the observed effects are short-term or novelty-driven, and that AI-generated titles could inadvertently lower overall engagement over time by prompting the recommender system to promote lower-quality videos, given that the absence of metadata (without GAI assistance) may signal low-quality or low-effort content. To address these concerns, we incorporate new longitudinal evidence to examine whether the impact of AI-generated titles persists over time. We collected an additional user-month-level dataset covering the period from August 2023 to August 2024. Following our experiment, the platform launched an AI-generated hashtag tool for all producers with more than 1,000 followers. To avoid confounding effects driven by this platform-wide rollout, we have restricted our analysis to producers in our experimental sample who had fewer than 1,000 followers. Leveraging this new dataset, we estimated an OLS regression model with robust standard errors specified as:

$$ValidWatch_{it} = \beta_1 Treat_{it} + \beta_2 Controls_{it} + \mu_i + \delta_t + \varepsilon_{it} \tag{P.2}$$

where $Treat_{it}$ indicates whether user i had access to AI-generated titles in our sample period, μ_i denotes user fixed effects and δ_t represents month fixed effects. Model estimation results are presented in Table P.3. We find that the coefficient of *Treat* is positive and significant, consistent

with the short-term results in Table 4. These findings suggest that the positive effect on video consumption is persistent rather than driven by short-term novelty.

Table P.3 Long-term Effect Analyses

Variables	*ValidWatch*
	(1)
Treat	**0.003*****
	(0.001)
Controls	Yes
Observations	12,681,558
R-squared	0.244

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Values in parentheses are robust standard errors.

P.3. Results for Viewership Diversity Analysis

For analyses reported in the main text (as explained in Section 5.1), we focused on the influence of AI-generated titles on video viewership. Here, we extend this analysis to explore their impact on viewership diversity. We first divided the total videos in our sample into two groups: videos produced by treatment group users and videos by control group users. We then calculated HHI based on valid watches to measure viewership concentration for each group, defined as:

$$\text{HHI} = \sum_{j=1}^{N} s_j^2 \tag{P.3}$$

where s_j represents the market share of video j within its respective group, calculated as:

$$s_j = \frac{V_j}{\sum_{j=1}^{N} V_j} \tag{P.4}$$

where V_j is the number of valid watches for video j, and the denominator is the total number of valid watches across all N videos within the group.

We used 1000 bootstrap iterations for each group to calculate HHI. In each iteration, we randomly sampled videos with replacement from each group to create a bootstrap sample and calculated the HHI for this sample using equations (P.3) and (P.4). This process yielded 1000 HHI values for both the treatment and control groups, enabling a bootstrap-based significance test to assess whether the difference in viewership concentration between the two groups was statistically significant. The results in Table P.4 show that the treatment group with access to AI-generated titles had a significantly lower HHI of 0.0002 compared to 0.0003 in the control group ($p < 0.05$), representing a 33% reduction in platform-level HHI. A lower HHI for the treatment group indicates that viewership of videos is more evenly distributed across videos in the treatment group than across those in the control group, reflecting greater diversity in viewership of videos.

Table P.4 Results of Platform-level HHI Analysis

Treatment Group	Control Group	Difference	P-value
(1)	(2)	(3)	(4)
0.0002	0.0003	-0.0001	< 0.05

Notes: We perform 1000 bootstrap iterations to compute the HHI and p-value.

P.4. Results for Alternative Operationalization of Variables

To examine the robustness of results, we employed alternative operationalizations for key variables in our analysis. Given that the observed positive effect of AI-generated titles may vary depending on how viewership is defined, testing these alternative definitions allows us to verify the consistency and generalizability of our results. First, for viewership, we used the number of watches ($Watch_{ij}$) and the number of likes ($Like_{ij}$) per video as alternative dependent variables. These measures capture different aspects of viewer engagement and may offer additional insights into how AI-generated titles impact a range of interactive forms of engagement. Results in Tables P.5 and P.6 are qualitatively consistent with our findings in Section 5.1, confirming that AI-generated titles positively impact users' content consumption and engagement.

Table P.5 Results Using Alternative Dependent Variables

Variables	*Watch*	*Like*
	(1)	(2)
Treat	**0.015*****	**0.021*****
	(0.005)	(0.004)
Controls	Yes	Yes
Observations	10,738,984	10,738,984
R-squared	0.313	0.297

Notes: $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.1$. Values in parentheses are robust standard errors clustered at the producer level.

Second, we considered two alternative measures of title similarity to validate the robustness of our findings. First, we used the Levenshtein distance, also known as the edit distance, to compute the similarity between AI-generated and posted titles as an alternative to cosine similarity used in Section 6.2. This metric calculates the minimum edits (e.g., insertions, deletions, substitutions) needed to transform one title into another, with higher distances indicating lower similarity. Using this alternative measure, we then replicated the analysis in Section 6.2. The positive coefficients of *Treat* and negative coefficients of the interaction term between *Treat* and *Similarity*, as shown in Tables P.7 and P.8, are qualitatively aligned with our main analysis.

Table P.6 Heterogeneous Results Using Alternative Dependent Variables

Variables	*Watch* (1)	*Like* (2)
Treat	0.017***	0.025***
	(0.005)	(0.004)
Treat × Utilitarian	**-0.035*****	**-0.056*****
	(0.013)	(0.009)
Controls	Yes	Yes
Observations	10,738,984	10,738,984
R-squared	0.313	0.297

Notes: ***p<0.01; **p<0.05; *p<0.1. Values in parentheses are robust standard errors clustered at the producer level. *Utilitarian* is absorbed by the video category dummies in *Controls* and therefore not reported in the table.

Table P.7 Model Estimation Results Using Levenshtein Algorithm (Matched Sample)

Variables	*ValidWatch* (1)	*WatchDuration* (2)
Treat	**0.251*****	**0.208*****
	(0.002)	(0.002)
*Treat * Similarity*	**-1.062*****	**-0.880*****
	(0.003)	(0.002)
Controls	Yes	Yes
Observations	3,885,089	3,885,089
R-squared	0.385	0.396

Notes: ***p<0.01; **p<0.05; *p<0.1. Values in parentheses are robust standard errors.

Table P.8 Results of Lexical Richness Analysis Using Levenshtein Algorithm (Matched Sample)

Variables	*LexicalDensity* (1)	*LexicalVariation* (2)	*Entropy* (3)
Treat	**0.031*****	**0.034*****	**0.030*****
	(0.0002)	(0.0002)	(0.001)
*Treat * Similarity*	**-0.263*****	**-0.322*****	**-0.464*****
	(0.0003)	(0.0002)	(0.001)
Controls	Yes	Yes	Yes
Observations	3,885,089	3,885,089	3,885,089
R-squared	0.634	0.689	0.802

Notes: ***p<0.01; **p<0.05; *p<0.1. Values in parentheses are robust standard errors.

Second, we employed GPT-5 to directly evaluate the semantic similarity between AI-generated and actual titles adopted by producers. Each pair of titles (an AI-generated title and its corresponding human-adopted title) was provided to GPT-5 with a prompt instructing it to output a continuous similarity score ranging from 0 to 1, where 0 indicates completely unrelated titles and

1 indicates identical titles. We then replicated our analyses with this alternative measure. Results in Tables P.9 and Table P.10 are qualitatively aligned with Tables 11 and 12 in our main analysis. Together, these alternative variable operationalizations indicate that our analyses are robust across different measures of consumption and title similarity.

Table P.9 Model Estimation Results Using LLM-Based Semantic Similarity (Matched Sample)

Variables	*ValidWatch* (1)	*WatchDuration* (2)
Treat	**0.263*****	**0.219*****
	(0.002)	(0.002)
*Treat * Similarity*	**-1.078*****	**-0.896*****
	(0.003)	(0.002)
Controls	Yes	Yes
Observations	3,885,089	3,885,089
R-squared	0.382	0.393

Notes: $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.1$. Values in parentheses are robust standard errors.

Table P.10 Results of Lexical Richness Analysis Using LLM-Based Semantic Similarity (Matched Sample)

Variables	*LexicalDensity* (1)	*LexicalVariation* (2)	*Entropy* (3)
Treat	**0.034*****	**0.030*****	**0.035*****
	(0.0002)	(0.0002)	(0.001)
*Treat * Similarity*	**-0.319*****	**-0.261*****	**-0.469*****
	(0.0002)	(0.0002)	(0.001)
Controls	Yes	Yes	Yes
Observations	3,885,089	3,885,089	3,885,089
R-squared	0.684	0.631	0.804

Notes: $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.1$. Values in parentheses are robust standard errors.